German Internal Migration: A Marriage Market Perspective *

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The most recent version of this paper is continuously updated and is available at https://ali-demir.org/files/Demir_JMP.pdf .

Abstract

This paper studies whether the large East–West migration after German reunification changed marriage and divorce in receiving West German districts. Using newly digitized district panels of divorces, marriages and births for 1985–2007, extending to 1980 for a subset, I compare districts within the same state by the share of their 1961 population that had moved from East Germany before the Berlin Wall. This share strongly predicts post-1991 arrivals. After reunification, divorce fell in proportion to the share: by about 2 percent of the mean per percentage point, and by 14 percent in the top relative to the bottom quintile, with no differential trends during the 1980s. The decline appears by 1994, a few years after the largest migration wave of 1989–1990; expressed per young arrival measured since 1991, it equals 0.027 fewer annual divorces per 1,000 residents. A placebo share of expellees without East German ties predicts neither migration nor divorce, and districts with stronger ties did not grow faster. Marriages do not rise, and the decline is far larger than the small induced change in sex ratios could produce. Fewer divorces without more marriages is the pattern predicted by rising costs of forming and dissolving households, not by a thicker partner market.

Keywords: Migration, divorce, marriage, shift-share, German reunification.
JEL Codes: J12, J61, R23, D10.

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1 Introduction

German reunification triggered a large and persistent wave of internal migration from East to West Germany that reshaped local demographic and economic conditions across West German districts (*Kreise* and *kreisfreie Städte*) (Fuchs-Schündeln & Schündeln, 2009; Glorius, 2010; Stawarz et al., 2020). Since reunification, East Germany experienced a large net population loss through internal migration to West Germany, exceeding 1.2 million people over the post-reunification period; gross East–West mobility was substantially larger (Rosenbaum-Feldbrügge et al., 2022; Stawarz et al., 2020). These flows were also strongly selective. Young and relatively educated women moved disproportionately, while return migration was more common at older ages (Fuchs-Schündeln & Schündeln, 2009; Kröhnert & Vollmer, 2012; Leibert, 2016). This demographic reallocation matters because marriage and divorce affect household resources and the incentives for marriage-specific investments, including children and within-household specialization (Becker et al., 1977; Bröckel & Andreß, 2015).

This paper therefore asks: did greater exposure to young East–West migration after German reunification affect marriage formation and dissolution in receiving West German districts? The direction of the effect is not obvious a priori, and the joint response of marriages, divorces, and court filing records helps distinguish between the competing mechanisms. First, young arrivals add singles to the local partner market, which lowers search frictions for unmarried residents (Becker et al., 1977) and improves outside rematching options for married spouses (South & Lloyd, 1995; Uggla & Andersson, 2018); a thicker partner market should therefore raise both marriages and divorces. Second, to the extent that the inflow is female-skewed, it weakens women's outside options relative to men's, which should lower wife-initiated relative to husband-initiated divorces (Angrist, 2002; Chiappori et al., 2002). Because both sexes moved in large numbers into populous western districts and the initial 1989–1990 surge was male-led, however, the induced change in local sex ratios per arrival is mechanically very small.

Arrivals raise local housing demand and rents (Saiz, 2007), increasing the costs of forming and dissolving households. Because cohabiting spouses share a single dwelling whereas divorce requires splitting one household into two separate residences, higher housing costs would make marital dissolution more expensive, lowering husband-, wife-, and jointly initiated divorces alike, while having a more muted effect on marriage formation as higher setup costs compete with incentives to pool rent. Finally, if arrivals displace incumbent residents to other districts (Borjas, 2006), local marriage, divorce, and birth counts scaled by fixed pre-reunification population would all fall proportionally without any change in behavior. Because district inflows are not observed jointly by age and sex before 2000, exposure is measured using total young arrivals aged 18–29 throughout.

The central empirical problem is that migrant destination choice is not random. East

German migrants may have sorted into economically dynamic, socially distinctive, or geographically advantaged West German districts, and those same districts may have experienced different family trajectories even in the absence of migration. To address this endogeneity, a shift-share instrument is utilised in the spirit of Card (2001). The shares represent a historical East-linked settlement share, denoting the proportion of each district's residents in 1961 who had relocated from East Germany prior to the construction of the Berlin Wall. The shift measures cumulative young East–West arrivals to sample districts in other West German states from 1991 onward per 1,000 East-linked residents recorded in those districts in 1961. The instrument thus predicts a district's exposure by distributing the aggregate inflow across districts in proportion to their historical ties to East Germany; Section 4 details its construction. The share component follows the logic of Burchardi and Hassan (2013), who use historically rooted East-linked settlement patterns as a proxy for social ties to East Germany. This approach is motivated by evidence that such ties shaped migrant destination choice after reunification (Dorner et al., 2016). Because all districts in a state face the same shift in a given year, identification rests on the historical shares: conditional on state-by-year fixed effects and predetermined characteristics, districts with stronger historical East ties must not have followed different marriage and divorce paths for reasons other than migration (Goldsmith-Pinkham et al., 2020). Because the shares are measured in the census of June 1961, shortly before the Berlin Wall halted East–West migration for 28 years, they are not contaminated by ongoing adjustment to recent inflows, a central weakness of past-settlement instruments (Jaeger et al., 2018). Two threats remain. First, regions with stronger East ties grew faster after reunification (Burchardi & Hassan, 2013). Second, districts near the former border, which had higher 1961 shares, gained market access and lost place-based support (Ehrlich & Seidel, 2018; Redding & Sturm, 2008). I assess the identifying assumption with event studies that use newly digitized divorce counts back to 1980, with distance-to-border-by-year fixed effects, with newly assembled district income data for 1989–2007, and with a placebo: the 1961 share of expellees who came to the West without first living in the Soviet zone, which predicts neither arrivals nor divorce.

The main finding is that divorce fell after reunification in districts with stronger historical ties to the East. Relative to 1985–1988, annual divorces after 1991 fell by 0.054 per 1,000 baseline residents for each percentage point of the 1961 East-linked share, or 2.1 percent of the mean; a Poisson model for the divorce counts gives 2.3 percent. Districts in the highest quintile of the share experienced a relative decline of approximately 0.35 annual divorces per 1,000 baseline residents compared with districts in the lowest quintile. The decline becomes larger with each successive quintile, and event studies on balanced panels reaching back to 1980 and 1982 show no differential trends before reunification. The instrumental-variable estimate scales this decline by measured arrivals. In the preferred specification, which includes district and state-by-year fixed effects, a

district receiving one additional young arrival per 1,000 residents had 0.027 fewer annual divorces per 1,000 residents; a one-standard-deviation increase in exposure corresponds to about 9 percent of the mean. The first stage is strong, and weak-instrument-robust confidence sets exclude zero. Conditioning on pre-reunification divorce levels instead of district fixed effects gives a somewhat larger estimate of 0.036. The estimate is similar or larger when shocks that vary with distance to the former inner-German border are absorbed, when district income is controlled for, when the shift is constructed differently, when extreme-exposure districts are excluded, when any single state is dropped, and when districts are weighted by population. Marriages do not rise in any specification. In the reduced form they fall by 1.1 percent per percentage point of the share, but the decline is not monotone across quintiles of the share, and with district fixed effects the instrumental-variable estimate is small and imprecise.

The timing of the response reflects when the migration took place. The decline emerged in the mid-1990s, a few years after the largest wave of 1989–1990, which district exposure misses because the migration matrices begin in 1991; per measured arrival, the estimate therefore falls from 0.118 in 1992–1996 to 0.034 in 2002–2007. When the arrival rate of the instrument, which requires only state totals, is extended to 1989–1990, cumulative predicted arrivals describe the year-by-year path of the decline better than a one-time post-reunification shift. Measured exposure also counts gross arrivals, and historical ties raised departures to the East by three quarters as much as arrivals. The instrumental-variable coefficient therefore rescales the reduced form by measured arrivals and is not a constant effect of each additional arrival. The decline does not differ between husband- and wife-initiated divorces, and it is far too large to result from the small change in local sex ratios that a moderately female-skewed inflow implies. Together with the absence of any increase in marriage, these results are inconsistent with a thicker partner market and do not support a sex-ratio channel. Births do not rise as much as they would if every arrival had stayed and had the fertility of West German residents of the same age, so the district data cannot establish whether the decline is specific to marital dissolution. Two explanations that work through the number of residents find little support: net migration with other West German districts did not fall as arrivals accumulated, which gives no indication that arrivals displaced incumbent residents, and departures to the East account for about a seventh of the decline. The pattern of fewer divorces without more marriages has the signs that higher costs of forming and dissolving households predict, but the district data cannot test this channel directly: the available housing measures describe the housing stock in 1987, not rent growth after 1989.

This paper contributes to three strands of the literature. First, it makes a data contribution: I assemble, to my knowledge for the first time, a harmonized district-level panel of divorce, marriage and birth counts for West Germany that covers 320 districts from 1985 to 2007 and reaches back to the early 1980s for 241 districts in six states,

digitized from printed publications of the state statistical offices, mapped to a stable 1991 district geography, and linked to annual district marriage counts compiled from official statistical publications. Second, the study makes a significant contribution to the extant literature on German internal migration after reunification. It demonstrates that the consequences of East-West mobility extend beyond labour-market adjustment and demographic redistribution, to include marital dissolution in the regions to which migrants are relocated (Fuchs-Schündeln & Schündeln, 2009; Glorius, 2010; Stawarz et al., 2020). Third, it contributes to the economics of marriage markets and to the literature on migration shocks and predetermined networks by showing that a large internal migration of young adults was followed by a decline in divorce rates and an absence of an increase in marriage rates in more exposed destination markets. This pattern is not consistent with the prediction that an increased number of potential partners will result in an increase in both marriage and divorce (Becker et al., 1977; South & Lloyd, 1995), and the divorce decline is far too large to operate through the induced change in local sex ratios (Angrist, 2002; Chiappori et al., 2002). The design combines the new panel with a shift-share instrument based on settlement shares recorded before the Berlin Wall was built, and tests for pre-reunification trends with the newly digitized data from the 1980s (Goldsmith-Pinkham et al., 2020; Jaeger et al., 2018).

The remainder of the paper is organized as follows. Section 2 reviews the historical background and develops the paper's testable hypotheses. Section 3 describes the data, sample construction, and descriptive evidence. Section 4 presents the empirical framework and identification strategy. Section 5 reports the main results, heterogeneity analyses, and mechanism evidence. Section 6 concludes.

2 Background

German reunification triggered one of the largest internal migration episodes in an advanced economy. The 18–24 age group accounted for roughly one third of net East–West migration losses between 1991 and 2006 (Glorius, 2010). Using official migration statistics for 1991–2012, Stauder (2018) shows that female overrepresentation in East–West migration is concentrated among the youngest adult movers. SOEP-based projections for 1991–2012 indicate that, by age 28, around 87,000 East German men and 239,000 East German women had moved West in connection with education. Micro-data further show that employed East–West migrants were less likely to be married and positively selected: in social insurance records only about 40% of employed East–West migrants in the mid-1990s were married compared to roughly 57% of East German non-migrants. Panel evidence indicates that individuals in partnerships are significantly less likely to move, while older single migrants are more likely to return to the East than to remain in the West (Brücker & Trübswetter, 2007; Fuchs-Schündeln & Schündeln, 2009). Over-

all, post-reunification East–West migration involved substantial flows of young adults of both sexes, with female overrepresentation particularly evident among education-related movers.

Figures 1 and 2 illustrate the aggregate timing and demographic composition of this migration episode, highlighting three features that shape the empirical design and the interpretation of the estimates.[1] Figure 1 plots the annual net East-West migration of German nationals by sex between 1985 and 2007, defined as moves from East to West Germany minus moves in the opposite direction. Prior to the opening of the Berlin Wall on 9 November 1989, institutional restrictions kept net East-West mobility below 30,000 moves per year. The border opening and formal unification on 3 October 1990 triggered an extraordinary, front-loaded surge: net migration rose to approximately 300,000 per year in both 1989 and 1990 before falling to 150,000 in 1991 and stabilising at a range of 20,000 to 70,000 per year between 1992 and 2007. The majority of net population transfers took place between late 1989 and 1992. Consequently, behavioural adjustments in receiving marriage markets should materialise primarily by the mid-1990s.

Second, Figure 1 shows that across the full 1985–2007 period, cumulative net East-West migration totaled 725,620 men and 808,914 women, implying an overall female share of 52.7%. During the peak surge of 1989–1990, male net migration (reaching roughly 175,000 in 1990) actually exceeded female net migration (roughly 135,000 to 140,000). Female overrepresentation emerged only from 1991 onward, when male net inflows dropped sharply while female net inflows remained between 20,000 and 60,000 per year. Thus, districts connected to East German migration networks absorbed a broad-based influx of both sexes during the initial unification shock, followed by moderately female-skewed net inflows in the mid-1990s.

Third, Figure 2 plots annual gross arrivals from East Germany by age group from 1991 onward, when district migration matrices by age begin (Section 3.2). Ages 18–29, whose arrivals define exposure in this paper, form the largest group in every year: their annual number fell from 101,000 in 1991 to 44,000 in 1994, rose to a second peak of 82,000 in 2001, and stood at 67,000 in 2007. Two patterns in Figure 2 shape the empirical interpretation. First, in the early 1990s, young arrivals were accompanied by substantial inflows of adults aged 30–49 (50,000 in 1991) and minors under 18 (55,000 in 1991), reflecting family migration; arrivals of minors and older adults subsequently fell more steeply, so the inflow became increasingly concentrated among young adults. Because historical settlement ties attracted both young singles and families during the initial transition, the instrument captures exposure to the broader reunification inflow scaled in

[1] Series are first aggregated at the state×year level and then summed to national totals. This figure restricts the sample to *German nationals*. For historical background and dates (Berlin Wall opening: 9 Nov 1989; formal unification: 3 Oct 1990; Two-Plus-Four Treaty entry into force: 15 Mar 1991), see, e.g., the chronology in the Federal Agency for Civic Education (bpb) and the Bundestag research note on the Two-Plus-Four Treaty.

units of young-adult arrivals. Second, comparing the large gross arrivals of 18–29 year-olds in the late 1990s and 2000s (Figure 2) with the much smaller net migration balances in those years (Figure 1) indicates substantial return and repeat migration in the later period. Accumulating gross arrivals from 1991 onward therefore tracks permanent net population additions more closely in the early 1990s than in the 2000s, when cumulative gross stocks increasingly reflect turnover.

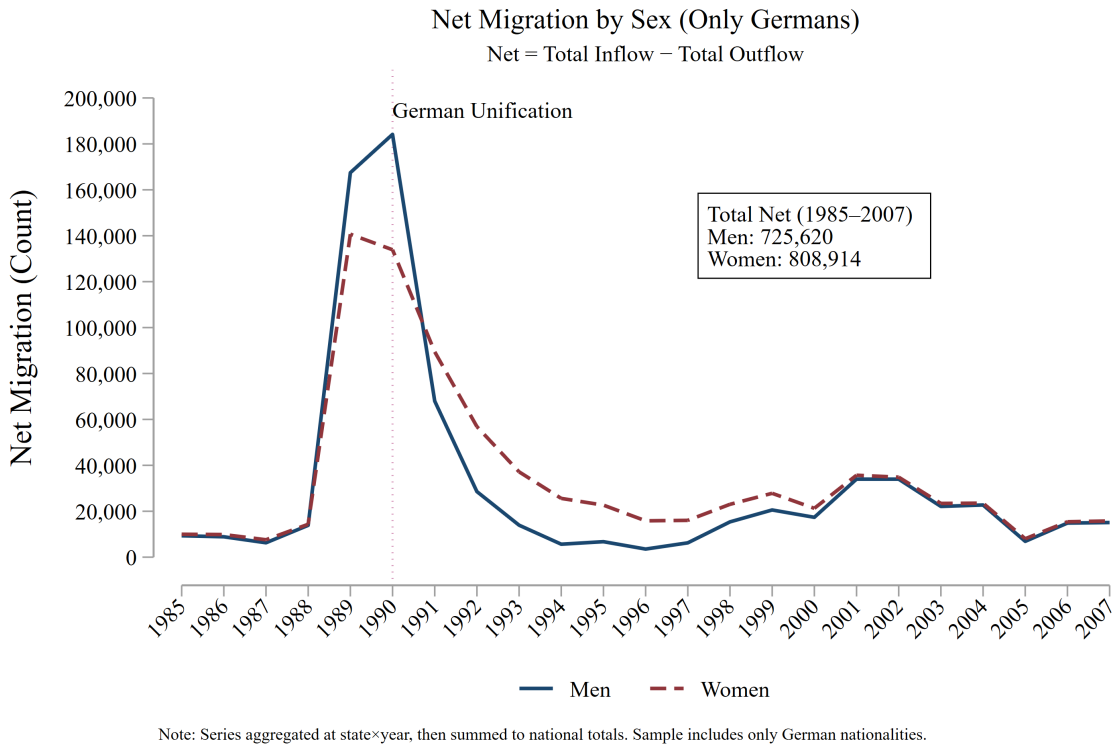


Figure 1: Net East–West internal migration by sex (German nationals only), 1985–2007

Notes: Net migration is defined as inflows from East Germany to West Germany minus outflows from West Germany to East Germany. Series are aggregated at the state×year level and summed to national East–West totals. The vertical line marks German unification.

Existing work on German reunification shows that historically rooted East–West ties shaped post-1989 economic behavior and spatial allocation. Burchardi and Hassan (2013) show that West German regions with stronger social ties to East Germany experienced faster post-reunification income growth, using historically determined East-linked settlement patterns and wartime destruction as a source of exogenous variation. Dorner et al. (2016) extend this logic to the migration of East German inventors and show that regions with stronger social ties to the East attracted more inventors after 1990. This literature makes it plausible that historically mediated East-links affected where migrants went after reunification. At the same time, spatial frictions and border shocks also mattered for regional adjustment near the former inner-German border (Redding & Sturm, 2008), and place-based interventions may have had persistent local effects (Ehrlich & Seidel, 2018). These considerations motivate the historical settlement component of the instru-

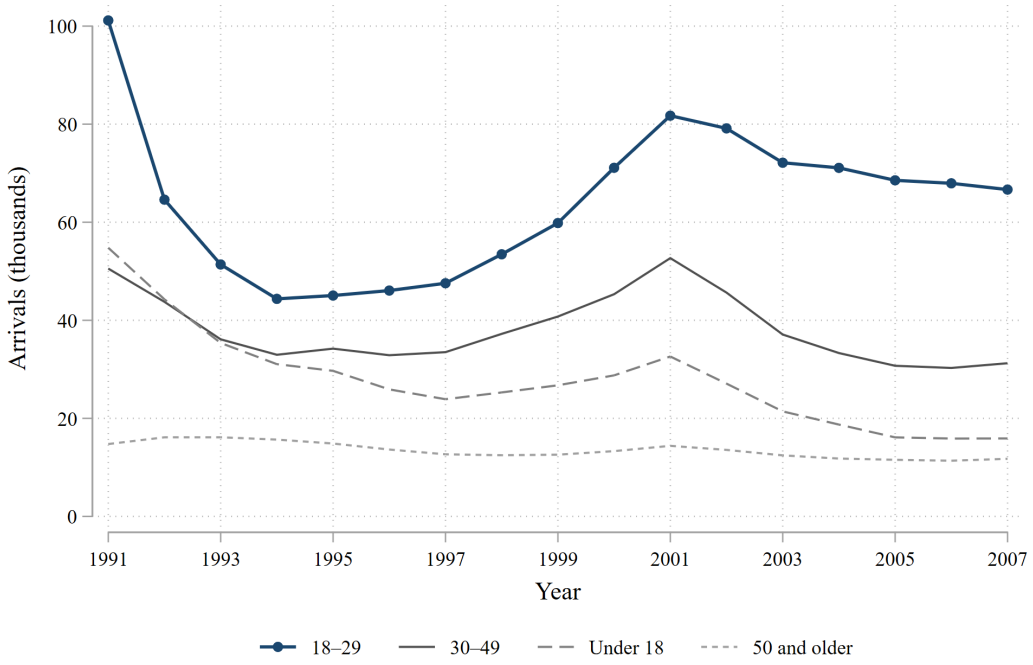


Figure 2: Arrivals from East Germany by age group, 1991–2007

Notes: Arrivals from the five eastern states (Berlin excluded) in West German districts outside the city-states, all nationalities, in thousands. Ages 18–29 as in the exposure measure (Section 3.2).

ment, while also highlighting potential violations of the exclusion restriction through other economic consequences of East–West ties and proximity to the former border.

2.1 Conceptual framework and hypotheses

I organize the possible effects of young in-migration on receiving districts with a standard model in which marriages form and dissolve according to their surplus (Becker et al., 1977; Peters, 1986). A single woman i and a single man j who meet marry, and an existing couple divorces, if

$$\text{marriage:} \quad V_{ij}^{M} - k^{F} > V_{i}^{S}(\mathcal{P}) + V_{j}^{S}(\mathcal{P}), \tag{1}$$

$$\text{divorce:} \quad V_{i}^{S}(\mathcal{P}) + V_{j}^{S}(\mathcal{P}) - k^{D} > V_{ij}^{M}, \tag{2}$$

where V_{ij}^{M} is the value of the match, which the partners can share; $V_{i}^{S}(\mathcal{P})$ is the value of being single for partner i given the local pool $\mathcal{P}$ of potential partners; and k^{F} and k^{D} are the costs of forming and of dissolving a joint household, such as the cost of one rather than two dwellings. The number of marriages also rises with the number of meetings, which increases with the number of singles in $\mathcal{P}$. Because the partners can share the surplus, whether a marriage dissolves depends only on the sum $V_{i}^{S} + V_{j}^{S}$ and not on which spouse files (Becker et al., 1977; Peters, 1986). Shocks that move the

outside options of the two spouses in opposite directions primarily change the division of the surplus (Chiappori et al., 2002); they affect dissolution only to the extent that the surplus cannot be fully reallocated, which would show up as a change in which spouse initiates divorce. Young East–West migration can enter equations (1) and (2) through three channels, which generate distinct predictions for marriages, divorces, and the filing composition (Table 1).

The first channel is a thicker partner market. Young arrivals add singles to the local pool, which raises the number of meetings and the value of being single for spouses who have access to this pool (South & Lloyd, 1995).

Hypothesis 1 (thicker partner market). Exposure to young arrivals raises both marriages and divorces in receiving districts.

The second channel is the sex composition of the inflow. Among young adults, more women than men moved from East to West, so the inflow makes women relatively less scarce, lowering women's and raising men's outside options (Angrist, 2002; Chiappori et al., 2002). Because both sexes move, however, the change in the local sex ratio per arrival is small. One additional young arrival per 1,000 residents changes the ratio of men to women aged 20–34 by approximately

$$\Delta R \approx \frac{(1-\phi) - R_0\,\phi}{F_0}, \tag{3}$$

where ϕ is the female share of young arrivals, R_0 is the baseline number of men per woman aged 20–34, and F_0 is the number of women aged 20–34 per 1,000 residents. With $\phi \approx 0.53$ at ages 18–25 (Stauder, 2018), $R_0 = 1.056$, and $F_0 \approx 115$ in the 1987 census data, $\Delta R \approx -0.0008$: even a large cumulative inflow moves local sex ratios only slightly.

Hypothesis 2 (sex composition). Exposure lowers divorce, more strongly for divorces initiated by wives, with effects bounded by the small induced change in local sex ratios.

The third channel works through the costs of forming and dissolving households. Inflows raise local housing demand and rents (Saiz, 2007), which increases both k^F and k^D: establishing a joint household and splitting one into two both become more expensive.

Hypothesis 3 (household costs). Exposure lowers both marriages and divorces, without a systematic difference between husband- and wife-initiated divorces.

Two non-behavioral mechanisms can also lower the outcomes, which are counts scaled by fixed pre-reunification population. If arrivals displace incumbent residents, as native internal migration responds to immigration in other settings (Borjas, 2006), local marriage and divorce counts fall without any change in behavior. And if fewer marriages form, fewer short-duration marriages are at risk of divorce later; because marriages that ended in divorce had lasted 13 to 14 years on average in 2000–2007 (Statistisches Bundesamt, 2025-06-26, 2025), this channel matters mainly in the later sample years.

Table 1: Predicted responses to young in-migration by channel

Channel	Marriages	Divorces	Wife- vs. husband-filed
H1: Thicker partner market	+	+	both rise
H2: Female-skewed composition	≈ 0	− (small)	wife-filed fall more
H3: Household costs	−	−	no difference
Non-behavioral mechanisms			
Displacement of residents	−	−	proportional
Fewer marriages at risk	—	− (later years)	proportional

Notes: Predicted signs for annual district counts scaled by fixed 1987 population. The divorce effect under H2 is bounded by the change in local sex ratios implied by equation (3). Under H1 and H3 the filing composition follows from the transferable-utility logic of equations (1) and (2).

The mechanism is not directly observed. I therefore use the joint response of marriages and divorces to discriminate between Hypotheses 1 and 3, and treat the filing-party decomposition, available for three states, as exploratory evidence on Hypothesis 2.

Finally, the effect may vary across local marriage markets. A given inflow should matter more where partner markets are thinner, where migration frictions are lower, or where historical East–West ties made post-reunification mobility more salient. I therefore examine heterogeneity by district type and by distance to the former inner-German border. These tests are descriptive: they show where the response is strongest but do not by themselves establish the behavioral mechanism.

3 Data

The analysis panel covers 320 West German districts (*Kreise*), 86 independent cities and 234 rural districts, on 1991 boundaries and observed annually from 1985 to 2007.[2]

3.1 Divorces, marriages and births

Divorce counts by district come from the annual reports on court-ordered dissolutions of marriage (*Gerichtliche Ehelösungen*) that each state statistical office publishes (Statistische Landesämter, 1981–2008). I digitized the printed tables by optical character recognition. The series begin in 1980 in Baden-Württemberg, Hesse, North Rhine-Westphalia and the Saarland, in 1981 in Schleswig-Holstein, in 1982 in Bavaria, and in 1985 in Lower Saxony and Rhineland-Palatinate, for which no earlier district tables are accessible in print or in digital form. All 320 districts are therefore observed from 1985, and two bal-

[2]The sample excludes the city-states (West Berlin, which the 1961 district data do not cover, Hamburg and Bremen) and four districts of Lower Saxony that cannot be matched across all sources on constant boundaries: the city and the district of Hannover, which merged into the Region Hannover in 2001, and the districts of Göttingen and Osterode am Harz, which merged in 2016.

anced panels reach further back: 241 districts in six states from 1982, and 130 districts in four states from 1980.

I checked the digitized counts against the scanned tables, using the totals that each table prints for administrative regions and for the state as accounting checks. The checks found three kinds of error in an earlier transcription: values copied from an adjacent year, counts assigned to a city instead of the surrounding district of the same name or the reverse, and misread digits. Most corrections concern Bavaria (1,296 district-year cells over 1982–2007); Appendix A.1 documents the checks.

Divorces are assigned to the district of residence that determines which family court is competent (Statistisches Bundesamt, 2010c). The reports also record who filed the petition: the husband, the wife, or both spouses. Section 5.4 uses this information for Lower Saxony, North Rhine-Westphalia and Baden-Württemberg, which I digitized for 1992–2007: 141 districts, with Lower Saxony missing in 1995 and 1997.

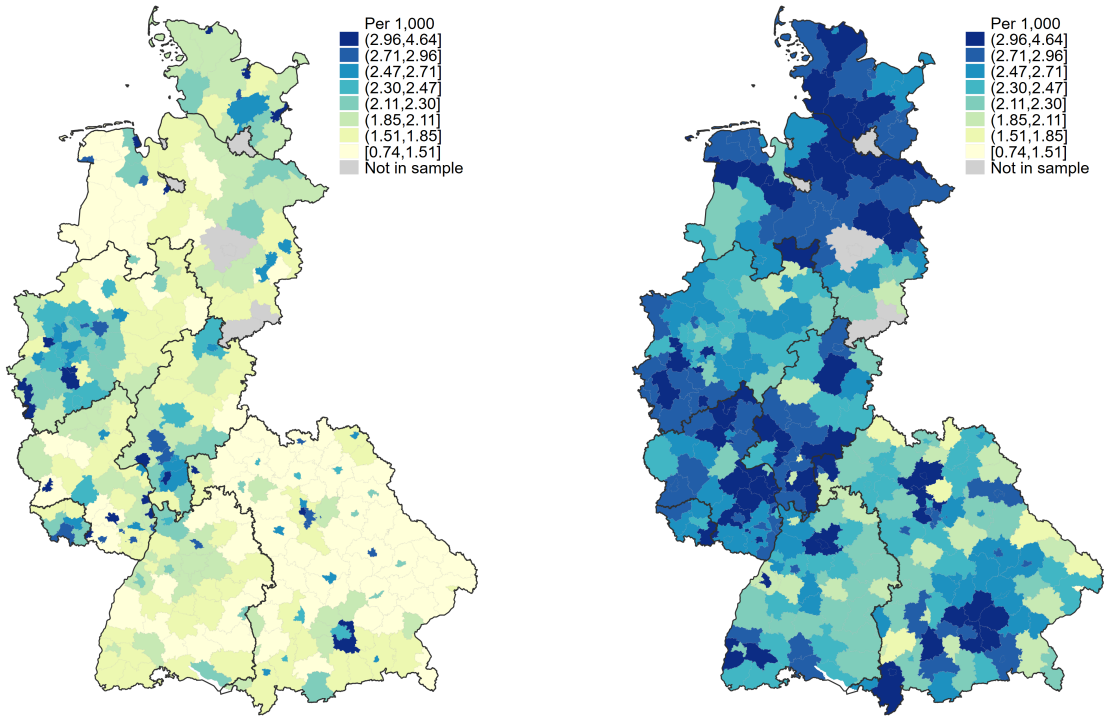
Marriages and live births, in total and to unmarried parents, come from the Federal Statistical Office: for 1985–1999 from the annual volumes *Gebiet und Bevölkerung* (Fachserie 1, Reihe 1), which I digitized in the same way, and for 2000–2007 from the digital district table N 50, *Eheschließungen, Geborene und Gestorbene nach Kreisen* (Statistisches Bundesamt, 1987–2001, 2000–2007). Births are assigned to the mother's district of residence (Statistisches Bundesamt, 2010a). Marriages are assigned to the registry office where the wedding takes place, wherever the spouses live (Statistisches Bundesamt, 2010b). Districts that attract weddings from elsewhere therefore record more marriages than their residents enter into. The clearest cases are Nordfriesland, Friesland and Wittmund, which include North Sea islands, and the spa town of Baden-Baden; Nordfriesland records up to 16 marriages per 1,000 residents a year, against a sample mean of 6.

All outcomes are expressed per 1,000 residents counted in the census of 25 May 1987:

$$y_{it} = 1000 \, \frac{E_{it}}{P_{i,1987}}, \tag{4}$$

where E_{it} is the number of divorces, marriages or live births in district i in year t, and $P_{i,1987}$ is the district's census population (Schmitt et al., 1994). Because the denominator is fixed, arrivals cannot lower the rates mechanically by enlarging the population. y_{it} is therefore a count scaled by a constant, not a hazard among married couples.

Figure 3 maps the divorce rate in 1987 and 2007. Divorce rose in 267 of the 320 districts, and in both years it was lower in Bavaria and Baden-Württemberg than in the rest of West Germany.



(a) 1987 (b) 2007

Figure 3: Divorces per 1,000 residents by district, 1987 and 2007

Notes: Annual divorces per 1,000 residents counted in the 1987 census, y_{it} in equation (4). Both panels use the same classes, the octiles of the pooled 1987 and 2007 values. In 2007, the four pairs of a city and its surrounding district whose published counts assign divorces to the wrong member of the pair (Appendix A.1) are shown at the pair's combined rate. Gray: districts outside the sample. Thick lines: state borders. District boundaries of 1 January 1998 (Bundesamt für Kartographie und Geodäsie, 1998).

3.2 Migration from East Germany

Arrivals come from the annual district-to-district migration matrices of the statistical offices (*Kreiswanderungsmatrix*), which are available in digital form from 1991 (Statistisches Bundesamt, 1991–2007). Each matrix counts registered moves of a main residence between districts by district of origin, district of destination and age group; from 2000 on, it also reports sex and nationality.[3]

For each district i and year τ, $M_{i\tau}$ is the number of arrivals aged 18–29 (the matrix's age groups 18–24 and 25–29) from districts in Brandenburg, Mecklenburg-Western Pomerania, Saxony, Saxony-Anhalt and Thuringia; moves from Berlin are not counted. Exposure is the cumulative inflow up to the previous year per 1,000 residents:

$$X_{it} = \frac{1000}{P_{i,1988}} \sum_{\tau=1991}^{t-1} M_{i\tau}, \tag{5}$$

[3]The unit is a move, not a person: someone who moves twice in a year is counted twice. Age refers to the end of the reporting year. Moves of foreign nationals are included.

where $P_{i,1988}$ is the district's population on 31 December 1988, the last year-end before the border opened (Statistisches Bundesamt, 1989), and $X_{it} = 0$ for $t \leq 1991$. X_{it} counts gross arrivals: neither return moves to the East nor onward moves within the West are netted out. Because the matrices begin in 1991, arrivals in 1989 and 1990 are not observed at the district level. The same matrices record moves from each West German district to the East and moves between West German districts. Section 5.4 uses them to measure departures to the East and net migration within West Germany.

The matrices do not report sex before 2000. The printed migration reports of the state statistical offices give the sex composition of arrivals from the eastern states at the state level (Statistische Landesämter, 1992–2008); I digitized them, and they enter only the calculation in Section 2.1.

3.3 The 1961 East-linked share

The census of 6 June 1961 recorded as a separate group the Germans who had lived in the Soviet zone or East Berlin on 1 September 1939 and moved to the West after the war, together with their children born since; expellees from the former eastern territories were counted separately (Ritter & Niehuss, 1987, p. 31). Burchardi and Hassan (2013, pp. 1230–1231) have no reliable regional data on this group, many of whose members never registered with the refugee authorities, and proxy its settlement with that of expellees who had lived in the Soviet zone before moving to the West. The census classifies residents by where they lived in 1939, not by registration, and so records the group itself. District counts of this group, R_i, come from the GESIS edition of the 1961 census district data (Blasius & Antoine, 1990), converted from the 565 districts of 1961 to 1991 boundaries (Appendix A.3). The share is

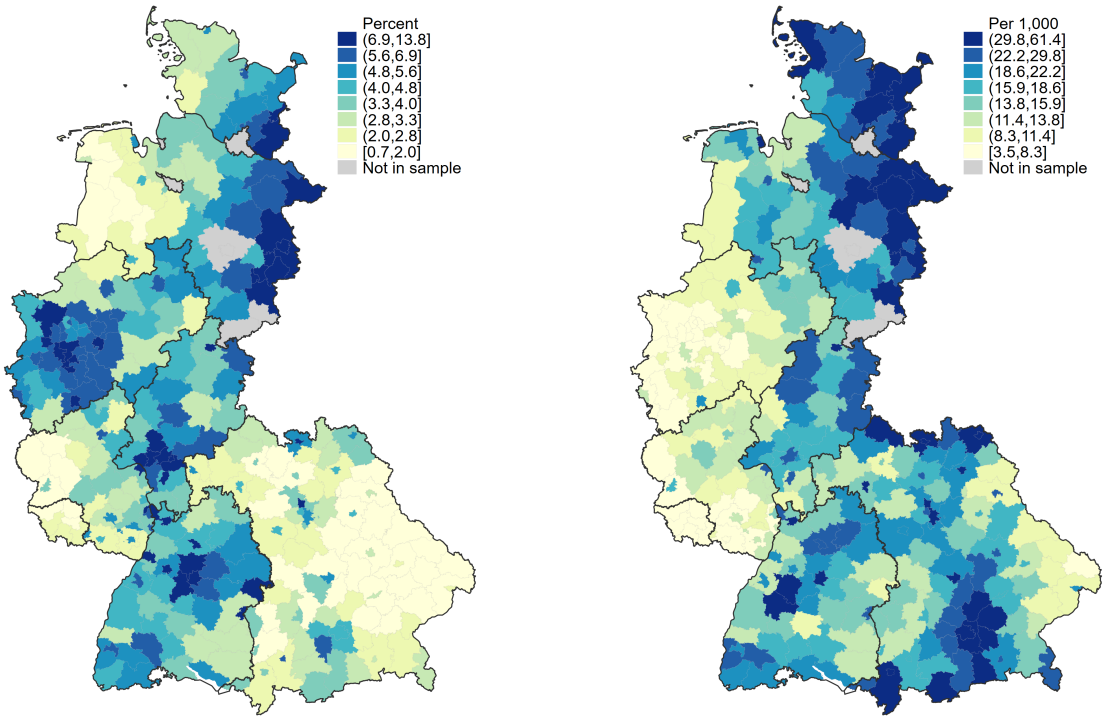
$$s_i = \frac{R_i}{P_{i,1961}}, \tag{6}$$

where $P_{i,1961}$ is the district's 1961 population on the same boundaries. In the seven states whose districts are all in the sample, the district counts summed to the state reproduce the published census shares, for example 3.1 percent in Bavaria, 5.7 percent in North Rhine-Westphalia and 6.3 percent in Hesse (Ritter & Niehuss, 1987, p. 31); Appendix Table A.1 reports all states.

Figure 4 maps the share next to exposure at the end of the sample, the arrivals of 1991–2006 ($X_{i,2007}$). Both vary considerably within states, which is the variation the estimates use: the preferred specification includes state-by-year fixed effects (equation (7)).

3.4 Baseline controls

The controls are district characteristics measured before reunification, all from the GESIS district file of the 1950–1987 censuses (Schmitt et al., 1994). From the 1987 census,



(a) 1961 East-linked share

(b) Arrivals aged 18–29, 1991–2006

Figure 4: The 1961 East-linked share and cumulative arrivals of young adults from East Germany, by district

Notes: Panel (a): the share s_i of equation (6), in percent of the 1961 population. Panel (b): arrivals aged 18–29 from the five eastern states in 1991–2006 per 1,000 residents of 1988, which is exposure $X_{i,2007}$ in equation (5). Classes are octiles of the 320 sample districts. Gray: districts outside the sample. Thick lines: state borders; with state-by-year fixed effects, the estimates compare districts within the same state. The correlation between the share and exposure is 0.25 across all districts and 0.47 within states. District boundaries of 1 January 1998 (Bundesamt für Kartographie und Geodäsie, 1998).

they are the share never married, log population density, the in-commuting rate, the unemployment rate, and the population share and the sex ratio (men per woman) at ages 20–34. From the 1970 census, they are the Catholic share of the population and women's share of the employed. The robustness and heterogeneity analyses also use each district's distance to the former inner-German border, computed from the federal administrative boundary files (Bundesamt für Kartographie und Geodäsie, 1998). Table 2 reports descriptive statistics.

4 Identification Strategy

This section sets out the estimating equations (Section 4.1), the instrument (Section 4.2), and the identifying assumption together with the threats to it (Section 4.3).

Table 2: Descriptive statistics

	Mean	SD	P10	P90
A. Outcomes per 1,000 residents of 1987, 1992–2007				
Divorces	2.52	0.53	1.86	3.17
Marriages	6.02	1.05	4.81	7.18
Live births	10.73	1.74	8.53	13.00
B. Migration and settlement				
Exposure, 1992–2007	9.66	7.93	2.33	19.55
Exposure in 2007	18.41	10.47	7.42	32.65
1961 East-linked share (%)	4.28	2.01	1.92	7.07
C. Baseline controls				
Share never married, 1987 (%)	38.61	2.59	35.40	41.70
Log population density, 1987	5.63	1.08	4.43	7.28
In-commuting rate, 1987 (%)	17.01	7.91	9.92	28.46
Unemployment rate, 1987 (%)	6.63	2.53	3.63	10.14
Population share aged 20–34, 1987 (%)	23.70	1.75	21.96	25.48
Sex ratio aged 20–34, 1987	1.06	0.04	1.01	1.11
Catholic share, 1970 (%)	50.11	28.69	8.68	86.53
Women's share of the employed, 1970 (%)	35.99	4.09	30.38	41.49

Notes: Divorces, marriages, live births and exposure in 1992–2007: 5,120 district-years (320 districts, 16 years). Exposure in 2007, the share and the controls: 320 districts. Outcomes are defined in equation (4), exposure (cumulative arrivals aged 18–29 per 1,000 residents of 1988) in equation (5), and the share in equation (6). Share never married: residents of all ages. Sex ratio: men per woman. P10 and P90: 10th and 90th percentiles.

4.1 Estimating equations

The preferred specification is

$$y_{it} = \beta X_{it} + \alpha_i + \delta_{s(i)t} + W_i'\gamma_t + \varepsilon_{it}, \tag{7}$$

where y_{it} is the outcome (equation (4)), X_{it} is exposure (equation (5)), α_i is a district fixed effect, $\delta_{s(i)t}$ is a fixed effect for the state $s(i)$ of district i in year t, and W_i is the vector of baseline controls (Section 3.4), whose coefficients γ_t differ by year. Because arrivals chose their destinations, exposure may be correlated with ε_{it}. I therefore estimate β by two-stage least squares, with first stage and reduced form

$$X_{it} = \pi Z_{it} + \alpha_i^{\pi} + \delta_{s(i)t}^{\pi} + W_i'\gamma_t^{\pi} + u_{it}, \tag{8}$$

$$y_{it} = \rho Z_{it} + \alpha_i^{\rho} + \delta_{s(i)t}^{\rho} + W_i'\gamma_t^{\rho} + e_{it}, \tag{9}$$

where Z_{it} is the instrument defined in equation (10), and each equation has its own fixed effects and control coefficients. With one instrument and one endogenous regressor, $\hat{\beta} = \hat{\rho}/\hat{\pi}$. The coefficient β is the change in annual events per 1,000 residents of 1987

when cumulative arrivals rise by one per 1,000 residents of 1988.

Exposure ends in the year before the outcome because divorce responds with a delay: a marriage can normally be dissolved only after the spouses have lived apart for a year (§ 1565 BGB). I use the same timing for marriages and births. Measuring exposure as a cumulative stock imposes that each arrival changes the outcome by the same amount in every later year, as the mechanisms in Section 2.1 imply. Section 5.1 relaxes this restriction by estimating the effect separately for three periods and examines it with an arrival rate extended to 1989–1990.

The tables report three specifications. The first includes only state-by-year fixed effects and compares outcome levels across districts of the same state and year. The second, a lagged-outcome specification, adds the baseline controls and the outcome's 1987 level with coefficients that are constant over time; it compares districts that had the same outcome before reunification. Both use 1992–2007. The third is equation (7). Its district fixed effects remove permanent differences between districts, and its year-specific control coefficients let districts with different baseline structures follow different paths, so that urban districts, for example, can follow a different divorce path from rural ones; Appendix B.4 reports the estimates without these coefficients. All controls are measured before reunification, so none can respond to the arrivals. This specification adds 1985–1988, when exposure is zero, so that the district effects are estimated from years before any arrivals. It omits 1989–1991, when arrivals were already large but exposure is recorded as zero (Section 3.2).

The lagged-outcome and fixed-effects specifications rest on different assumptions: that districts with the same 1987 outcome are comparable, and that districts would have followed parallel paths once permanent differences are removed. In a simpler setting with a binary treatment, Ding and Li (2019) show that the two approaches bracket the true effect; I report both and read the range between them as the plausible range of the effect. I prefer the fixed-effects specification because its identifying assumption can be tested with pre-reunification data (Section 4.3) and because it removes all permanent differences between districts, not only those captured by the 1987 outcome, whose association with later divorce changes over time (Section 5.3).

Standard errors are clustered by district. Because the model is just identified, the Anderson–Rubin test of $\beta = b_0$ is the test that Z_{it} has no effect in the reduced form with $y_{it} - b_0 X_{it}$ as the outcome. The values of b_0 that this test does not reject form a confidence set that remains valid when the instrument is weak (Anderson & Rubin, 1949). I report these sets together with the Kleibergen–Paap first-stage F statistic (Kleibergen & Paap, 2006).

4.2 The instrument

The instrument predicts each district's arrivals from its historical ties and from the rate at which such ties attracted arrivals elsewhere:

$$Z_{it} = s_i \sum_{\tau=1991}^{t-1} g_{-s(i),\tau}, \qquad g_{-s,\tau} = 1000 \, \frac{\sum_{j \notin s} M_{j\tau}}{\sum_{j \notin s} R_j}, \tag{10}$$

where s_i is the 1961 East-linked share (equation (6)), $M_{j\tau}$ is the number of young arrivals in district j in year τ (Section 3.2), R_j is district j's 1961 count of East-linked residents, and the sums run over the sample districts outside state s. As with exposure, $Z_{it} = 0$ for $t \leq 1991$.

The arrival rate $g_{-s,\tau}$ measures young East–West arrivals to sample districts in other West German states per 1,000 East-linked residents recorded in those districts in 1961. Z_{it} is therefore the exposure district i would have accumulated if its arrivals had followed its ties at the rate observed elsewhere, in the spirit of instruments built on earlier immigrant settlement (Card, 2001). If arrivals had been allocated exactly in proportion to ties, the first-stage coefficient π would equal $P_{i,1961}/P_{i,1988}$, about 0.9 on average, which gives the first-stage estimate a natural scale.

For each district, the aggregate arrival rate is calculated using sample districts in other West German states. Arrivals to the district's own state therefore do not enter this component of the instrument. State-by-year fixed effects absorb shocks common to districts within the same state and year. The robustness analysis also considers rates calculated from all other sample districts, including those in the same state, and from the full sample (Section 5.2).

Within a state and year, all districts use the same cumulative arrival rate calculated from the other West German states. With state-by-year fixed effects, the instrument therefore varies across districts only through s_i. The cumulative arrival rate determines the timing and scale of the comparison, while the historical share determines which districts are compared. Identification consequently rests on the historical shares, as in Goldsmith-Pinkham et al. (2020). The eight state-specific arrival-rate series largely reflect the same aggregate migration episode and therefore do not provide the many independent shocks required for an identification argument based on shock variation (Borusyak et al., 2025).

Instruments based on past settlement can also confound the response to new arrivals with continuing adjustment to earlier inflows from the same origin (Jaeger et al., 2018). Here the inflows that formed s_i ended when the Wall closed the border in 1961, 28 years before the new inflows began. Adjustment still under way in the 1980s would appear in the pre-reunification tests of Section 5.3.

4.3 Identifying assumption and threats

Because the instrument varies within a state and year only through s_i, the reduced form (9) is a restricted version of the event study

$$y_{it} = \sum_{k \neq 1988} \theta_k \, s_i \, \mathbf{1}[t = k] + \alpha_i + \delta_{s(i)t} + W_i' \gamma_t + \varepsilon_{it}, \tag{11}$$

where s_i is measured in percentage points and θ_k is the difference in the outcome in year k, relative to 1988, per percentage point of the share. The reduced form restricts θ_k to be proportional to the cumulative arrival rate in other West German states up to year $k-1$, apart from small differences in that rate between states: zero through 1991 and increasing with cumulative arrivals thereafter.

A one-parameter summary of equation (11) replaces the year-specific coefficients by a single change after reunification:

$$y_{it} = \theta \, s_i \, \mathbf{1}[t \geq 1992] + \alpha_i + \delta_{s(i)t} + W_i' \gamma_t + \varepsilon_{it}, \tag{12}$$

estimated on 1985–1988 and 1992–2007, where θ is the change in the outcome after reunification, relative to 1985–1988, per percentage point of the share. Unlike the reduced form (9), it does not impose that the response grows with cumulative arrivals.

The identifying assumption is that, without the arrivals, outcomes would have evolved alike in districts of the same state with different shares, given the controls:

$$\mathbb{E}\big[\, y_{it}(0) - y_{i,1988}(0) \,\big|\, s_i, W_i, s(i) \big] = \mathbb{E}\big[\, y_{it}(0) - y_{i,1988}(0) \,\big|\, W_i, s(i) \big], \tag{13}$$

where $y_{it}(0)$ is the outcome district i would have had in year t without arrivals from the East. Before 1989, $y_{it}(0) = y_{it}$, so the assumption implies $\theta_k = 0$ in every pre-reunification year, which Section 5.3 tests. After 1991, it is the exclusion restriction: arrivals must be the only channel through which historical ties changed outcomes. Pre-reunification data cannot test this part. The main threats are the following.

1. *Slow-moving differential trends.* Districts with stronger ties could have been on gradually diverging paths that a pre-period of four to nine years cannot detect. Section 5.3 reports how large a trend the tests could miss.

2. *Other effects of East ties.* Regions with stronger social ties to the East grew faster after reunification (Burchardi & Hassan, 2013), and ties may also have revived contact with relatives across the former border. Either could change family outcomes directly; Section 5.3 shows that the first is absent within states. Births, which arrivals should change only through their own fertility, provide a partial check: in the preferred and lagged-outcome 2SLS specifications they do not differ significantly

from that benchmark, and in the share reduced form they fall significantly below it only if every gross arrival is given at least 43 percent of the fertility of West German residents (Section 5.4). The check therefore does not exclude a broader effect of ties on family events. Ties also raised departures to the East, which lowers the number of residents at risk; Section 5.4 shows that this accounts for about a seventh of the decline.

3. *Shocks near the former border.* Districts close to the border gained market access after reunification (Redding & Sturm, 2008) and lost place-based subsidies when support for the former *Zonenrandgebiet* was scaled back after 1991 and ended in 1994 (Ehrlich & Seidel, 2018). To the extent that ties are stronger near the border, these shocks load on s_i. Section 5.2 adds bins of distance to the border interacted with year indicators.

4. *Unmeasured early arrivals.* If the arrivals of 1989 and 1990 also followed historical ties, the reduced form contains their effect on later outcomes while measured exposure does not, and β overstates the effect per arrival. Section 5.1 extends the arrival rate of the instrument to these years with state totals.

5. *Spillovers between districts.* Marriage markets extend beyond district borders, so arrivals can affect neighbouring districts. Spillovers into districts with lower shares bias the estimate toward zero. Correlated shocks across neighbouring districts are handled by clustering at the level of the 81 spatial planning regions (Section 5.2).

Three features of the design shape the interpretation of β. First, it is a local estimand: it averages the effects of arrivals whose destinations followed historical ties, weighting districts by how strongly their arrivals responded to those ties. Second, the outcomes count events per baseline resident (Section 3.1), so β combines changes in the rate at which couples marry or divorce with changes in the number of people at risk. If arrivals displaced married residents to other districts (Borjas, 2006), divorce counts would fall without any change in the divorce hazard. Section 5.4 examines this with migration between West German districts. Third, the arrivals' own events enter the counts. For divorce this contribution is positive but small, because young arrivals rarely divorce, so a negative β understates the decline among couples already living in the district. For marriages and births, the arrivals' own rates provide a benchmark against which the estimates can be judged.

5 Results

I begin by assessing whether the instrument defined in (10) predicts migration exposure and by reporting the reduced-form relationship between the instrument and the two

outcomes. Throughout the section, the tables report the three specifications defined in Section 4.1: state-by-year fixed effects only, the lagged-outcome specification, and the preferred specification.

Table 3 reports two first-stage exercises. Columns (1)–(3) regress annual inflow intensity, young East–West arrivals per 1,000 baseline residents in a given year, on the instrument. This is a construction check: it shows that the instrument predicts the flow of new arrivals and not only their accumulated stock. Columns (4)–(6) report the first stage for lagged cumulative exposure, the endogenous regressor in the 2SLS estimates.

Table 3: First-stage evidence for the instrument

	Construction check			Main first stage		
	Annual young inflow intensity			Cumulative exposure $_{t-1}$		
	(1)	(2)	(3)	(4)	(5)	(6)
Instrument	0.0608***	0.0564***	0.0418***	0.4553***	0.5085***	0.5483***
	(0.0071)	(0.0077)	(0.0074)	(0.0751)	(0.0817)	(0.1048)
State×year fixed effects	✓	✓	✓	✓	✓	✓
Baseline controls		✓			✓	
District fixed effects			✓			✓
Baseline controls×year			✓			✓
Pre-Years (1985–1988)			✓			✓
Observations	5,120	5,120	6,400	5,120	5,120	6,400
R^2	0.4596	0.5082	0.8657	0.6908	0.7197	0.9122
Instrument F statistic	72.56	53.30	32.33	36.72	38.71	27.39

Notes: Columns (1)–(3) regress annual inflow intensity, arrivals from East Germany aged 18–29 in year t per 1,000 residents of the fixed 1988 population, on the instrument. Columns (4)–(6) regress lagged cumulative exposure, the endogenous regressor in the 2SLS estimates, on the instrument. The instrument is the one-year-lagged shift-share predictor described in Section 4. Columns (1), (2), (4), and (5) use 1992–2007; columns (3) and (6) add 1985–1988, when exposure and the instrument are zero by construction, and exclude 1989–1991. Columns (2) and (5) include the baseline controls (Section 3.4) and the 1987 divorce rate; columns (3) and (6) interact the baseline controls with year indicators. Standard errors clustered by district in parentheses; the F statistic is the cluster-robust test of the instrument. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The instrument predicts exposure in every specification. In the preferred specification, one additional predicted arrival per 1,000 residents in 1961 raises measured cumulative arrivals by 0.55 per 1,000 baseline residents, with a first-stage F statistic of 27.4; the corresponding coefficients in the other specifications are 0.46 and 0.51. The instrument also predicts annual inflows, with F statistics between 32 and 73. These results establish relevance, not validity. Whether historical East-linked settlement shares are unrelated to other determinants of marriage and divorce is assessed in Section 5.3.

Table 4 reports the reduced form. Because the model is just identified, each 2SLS estimate below equals the ratio of the reduced-form coefficient to the first-stage coefficient, so the reduced form already determines the sign and significance of the IV results.

For divorce (Panel A), the reduced form is positive with state-by-year fixed effects only

Table 4: Reduced-form relationship between the instrument and the outcomes

	State-year FE (1)	Lagged outcome (2)	Preferred (3)
Panel A. Divorces per 1,000 residents			
Instrument	0.0151***	−0.0182***	−0.0150***
	(0.0039)	(0.0038)	(0.0050)
Panel B. Marriages per 1,000 residents			
Instrument	−0.0482***	−0.0289***	−0.0117
	(0.0111)	(0.0104)	(0.0106)
State × year fixed effects	✓	✓	✓
Baseline controls		✓	
District fixed effects			✓
Baseline controls × year			✓
Pre-reunification years (1985–1988)			✓
Observations	5,120	5,120	6,400

Notes: The dependent variables are annual divorces (Panel A) and marriages (Panel B) per 1,000 residents of the fixed 1987 population. The regressor is the one-year-lagged shift-share instrument. Samples and controls are as in Table 3; in Panel B, column (2) uses the 1987 marriage rate in place of the 1987 divorce rate. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and negative once permanent differences between districts are taken into account, either through the 1987 divorce level (−0.018) or through district fixed effects (−0.015). The positive coefficient in column (1) therefore reflects that districts with stronger historical East ties had persistently higher divorce levels; the event studies below show that these differences were stable before reunification. For marriage (Panel B), the reduced form is negative in all three columns, but in the preferred specification it is small and not statistically significant (−0.012, s.e. 0.011).

5.1 Main findings

Table 5 reports the change in the share gradient after reunification, θ in equation (12). Relative to 1985–1988, annual divorces fell by 0.054 per 1,000 residents for each percentage point of the 1961 share (s.e. 0.010), which is 2.1 percent of the sample mean; a difference of one standard deviation in the share (2.0 points) corresponds to 4.3 percent. A Poisson model for the divorce counts with the same fixed effects and controls, which measures proportional changes and does not depend on the denominator, gives a fall of 2.3 percent per percentage point (s.e. 0.5) and no pre-reunification pattern ($p = 0.43$). Marriages fall by 0.067 per percentage point (s.e. 0.032), or 1.1 percent of their mean; the end of this subsection shows that this decline is not robust. Column (4) reports births, to which Section 5.4 returns.

Table 6 reports OLS and 2SLS estimates for divorce (Panel A) and marriage (Panel B).

Table 5: Change after reunification by historical East-linked share

	Divorces	Divorces, Poisson	Marriages	Births
	(1)	(2)	(3)	(4)
1961 share × post	−0.0539***	−0.0233***	−0.0667**	−0.0714
	(0.0103)	(0.0046)	(0.0320)	(0.0508)
Percent of the mean per percentage point	−2.1	−2.3	−1.1	−0.7
Arrivals' own fertility (benchmark)				0.060
Pre-reunification coefficients (p)	0.26	0.43	0.52	0.89
Observations	6,400	6,400	6,400	6,400

Notes: Estimates of θ in equation (12): the coefficient of the 1961 East-linked share, in percentage points, interacted with an indicator for 1992–2007, in the preferred specification (district fixed effects, state-by-year fixed effects, baseline controls interacted with year indicators) on 1985–1988 and 1992–2007; 320 districts. Columns (1), (3) and (4): annual events per 1,000 residents of 1987. Column (2): Poisson pseudo-maximum-likelihood regression of the annual number of divorces with the same fixed effects and controls; the coefficient is a proportional change, so −0.0233 is −2.3 percent. The benchmark in column (4) is the change in births if every measured arrival stayed and had the fertility of West German residents: 0.045 births per resident aged 20–34 per year times the increase in cumulative young exposure per percentage point of the share after reunification, 1.32 (Appendix B.4). The p-value tests that the coefficients for 1985–1987 are jointly zero in equation (11), in column (2) in its Poisson version. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The dependent variables are annual divorces and marriages per 1,000 residents of the fixed 1987 population; exposure is cumulative young arrivals per 1,000 residents of the fixed 1988 population.

Table 6 scales the reduced form by measured exposure. In the preferred specification, the 2SLS coefficient in column (6) is −0.027: a district receiving one additional young arrival per 1,000 residents had 0.027 fewer annual divorces per 1,000 residents. The first stage is strong (Kleibergen–Paap $F = 27.4$), and the Anderson–Rubin confidence set, which remains valid under weak instruments, excludes zero. The standard deviation of cumulative exposure over 1992–2007 is 7.9 arrivals per 1,000 residents, so a one-standard-deviation increase corresponds to 0.22 fewer divorces per 1,000, about 9 percent of the sample mean of 2.52.

The lagged-outcome specification in column (5) yields a somewhat larger decline of 0.036, about 11 percent of the mean per standard deviation; following Section 4.1, I read the range between the two estimates as the plausible range of the effect. With state-by-year fixed effects alone (column 4), the coefficient is positive, reflecting the persistent level differences discussed above; this specification is not informative about the effect of exposure.

The OLS estimates are close to zero in all specifications, including −0.005 in the preferred one. Two forces plausibly bias OLS toward zero. Migrants are likely to have moved toward economically dynamic districts in which divorce would otherwise have risen, and exposure is measured as gross arrivals, which overstates the number of migrants who

Table 6: Young East–West migration exposure, divorce, and marriage

	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Divorces per 1,000 residents						
Cumulative exposure $_{t-1}$	0.003	−0.003	−0.005*	0.033***	−0.036***	−0.027***
	(0.003)	(0.003)	(0.003)	(0.011)	(0.008)	(0.009)
Kleibergen–Paap F				36.72	38.71	27.40
Panel B. Marriages per 1,000 residents						
Cumulative exposure $_{t-1}$	0.010	0.009	0.007	−0.106***	−0.056**	−0.021
	(0.013)	(0.009)	(0.008)	(0.031)	(0.022)	(0.020)
Kleibergen–Paap F				36.72	37.44	27.40
State × year fixed effects	✓	✓	✓	✓	✓	✓
Baseline controls		✓			✓	
District fixed effects			✓			✓
Baseline controls × year			✓			✓
Pre-Years (1985–1988)			✓			✓
Observations	5,120	5,120	6,400	5,120	5,120	6,400

Notes: The dependent variables are annual divorces (Panel A) and marriages (Panel B) per 1,000 residents of the fixed 1987 population. Exposure is cumulative arrivals from East Germany aged 18–29 between 1991 and $t-1$ per 1,000 residents of the fixed 1988 population. In the 2SLS columns, exposure is instrumented with the one-year-lagged shift-share predictor described in Section 4. Columns (1)–(2) and (4)–(5) use 1992–2007. Columns (3) and (6) add 1985–1988, when measured exposure and the instrument are zero by construction, and exclude the transition years 1989–1991. Columns (2) and (5) include the baseline controls (Section 3.4) and the 1987 level of the outcome (the divorce rate in Panel A, the marriage rate in Panel B); columns (3) and (6) interact the baseline controls with year indicators. Standard errors clustered by district in parentheses. Anderson–Rubin 95% confidence sets, which are robust to weak instruments: for divorces, $[-0.055, -0.022]$ in column (5) and $[-0.050, -0.010]$ in column (6); for marriages, $[-0.109, -0.016]$ and $[-0.065, 0.018]$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

remained and attenuates OLS. The 2SLS estimate is also local: it is identified from districts whose exposure was shaped by historical East ties.

The magnitudes should be read with three qualifications. At the 2007 mean of cumulative exposure, 18.4 arrivals per 1,000 residents, the preferred estimate implies about 0.50 fewer divorces per 1,000 residents relative to a district without exposure, or a fifth of the sample mean. First, measured exposure begins in 1991 and omits the 1989–1990 surge. If those arrivals followed the same networks, the estimates are effects per unit of measured post-1991 exposure and overstate the effect per actual arrival. Second, the size of the coefficient depends on which arrivals are counted. Arrivals of all ages are 2.3 times as numerous as young arrivals (43.1 against 18.4 per 1,000 residents by 2007); with them as the measure of exposure, the coefficient is −0.013 (s.e. 0.005; first-stage $F = 20.7$). Net of departures to the East, young arrivals respond to the instrument little more than a third as strongly as gross arrivals (0.21, s.e. 0.06, against 0.55), and the coefficient per net young arrival is −0.073 (s.e. 0.031; $F = 13.0$; Appendix Table B.13). The reduced

form is the same in all three cases; the 2SLS coefficient rescales it. Third, as shown below, most of the decline is in place by the mid-1990s, so it does not grow in proportion to the arrivals measured from 1991 onward.

Figure 5 reports the corresponding reduced-form event study for divorce: coefficients on the 1961 share interacted with year indicators, relative to 1988. Before reunification, the coefficients are small and jointly insignificant ($p = 0.26$), and the transition years 1989–1991 show no response. The coefficients turn negative from 1994: about −0.04 per percentage point of the share in 1994–1997, about −0.02 in 1998–2001, and between −0.06 and −0.07 in 2002–2005.

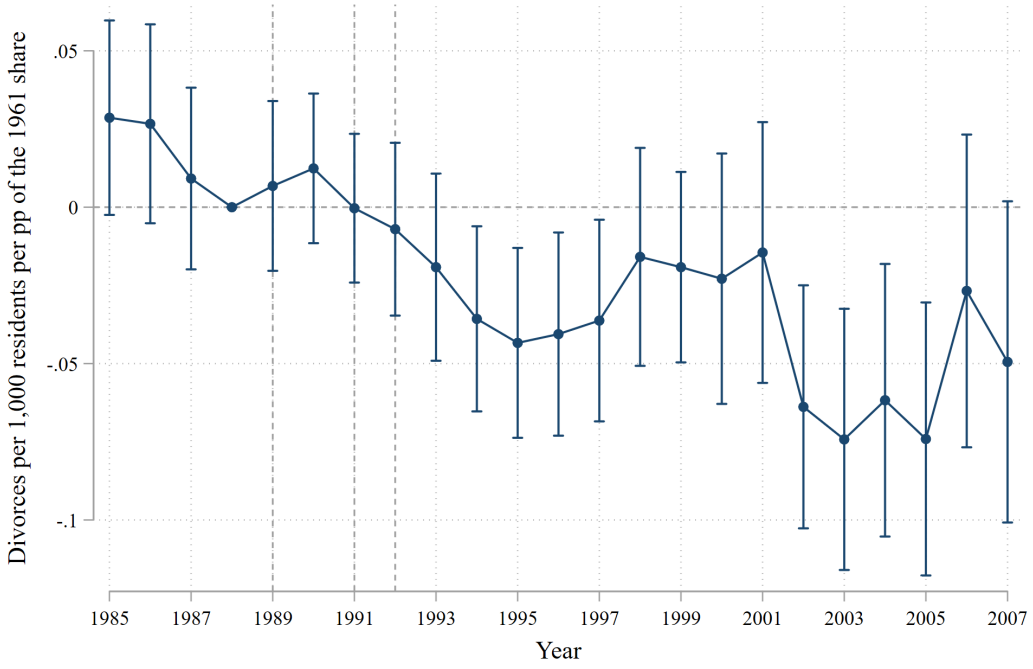


Figure 5: Event study of divorce by historical East-linked settlement share, 1985–2007

Notes: The figure plots the coefficients θ_k of equation (11) for divorces, with the share in percentage points; 320 districts, 1985–2007. Bars show 95% confidence intervals based on standard errors clustered by district. Vertical lines mark 1989, 1991, and 1992; 1989–1991 are transition years in which exposure is not measured.

The decline increases with the historical share across the whole distribution. Relative to districts in the bottom quintile of the 1961 share, divorce fell after reunification by 0.09 per 1,000 residents in the second quintile, by 0.24 and 0.32 in the third and fourth, and by 0.35 in the top quintile, about 14 percent of the sample mean (Appendix Table B.1). Because these quintile comparisons do not rely on a linear functional form, they also show that the estimates are not driven by a few districts with extreme shares.

The decline emerges early and grows far less than measured exposure. Against the 1985–1988 mean, the share gradient is −0.045 (s.e. 0.010) in 1992–1996, −0.038 (s.e. 0.012) in 1997–2001 and −0.074 (s.e. 0.015) in 2002–2007 (Appendix Table B.3). The

three differ ($p = 0.02$), but the first already amounts to 60 percent of the last, whereas cumulative arrivals rise throughout the period. When a common post-1991 shift in the gradient and the instrument enter together, the shift is −0.038 (s.e. 0.013), and the instrument, which grows with cumulative arrivals, adds −0.008 (s.e. 0.006). Scaled by measured exposure, the same reduced form therefore yields a 2SLS coefficient that falls from −0.118 (s.e. 0.041) in 1992–1996 to −0.035 (s.e. 0.014) in 1997–2001 and −0.034 (s.e. 0.010) in 2002–2007; the lagged-outcome specification shows the same pattern (Appendix Table B.2), and within districts over 1992–2007 the effect of exposure is small and imprecise (−0.015, s.e. 0.010; Appendix Table B.7). The IV coefficient on exposure measured from 1991 therefore scales a decline that is largely in place by the mid-1990s; it is not the marginal effect of an additional arrival within a district over time. The timing is consistent with a response to the inflows of the early 1990s, given that German divorce law generally requires a year of separation before a divorce can be granted. The largest of these inflows, in 1989 and 1990, is missing from measured exposure but not from the arrival rate: because $g_{-s,\tau}$ in equation (10) uses only arrivals in other states, state totals suffice to extend it to these years (Figure 1; Appendix A.2). The 1989–1990 wave then adds about as much to the cumulative arrival rate as the arrivals of 1991–1995 combined. Fitted by minimum distance to the sixteen post-reunification coefficients of the event study, cumulative predicted arrivals since 1989 describe their path better than a one-time shift ($J = 24.6$ against 36.6, each with 15 degrees of freedom) and as well as a shift combined with cumulative arrivals since 1991 ($J = 24.6$ with 14 degrees of freedom). This one-parameter path is not rejected at the 5 percent level ($p = 0.056$), and counting arrivals of all ages, which requires no assumption about the age of the 1989–1990 arrivals, gives the same result ($p = 0.058$; Appendix Table B.4 and Figure B.1).

Panel B of Table 6 shows no significant increase in marriages in any specification. With state-by-year fixed effects only and in the lagged-outcome specification, marriages fall, by 0.106 and 0.056 per arrival; the Anderson–Rubin set for the latter excludes zero. In the preferred specification, the estimate is −0.021, and its Anderson–Rubin set includes zero.[4] The marriage event study, reported in Appendix B.2, shows no differential trend before reunification ($p = 0.52$) and negative coefficients afterward, but marriages do not decline monotonically across quintiles of the share (Appendix Table B.1). The marriage decline is therefore not robust; the robust finding is the absence of an increase.

[4]Four districts with exceptionally high marriage rates relative to their population—Nordfriesland, Friesland, and Wittmund, which include North Sea islands, and Baden-Baden—are popular wedding locations. Excluding them reduces the lagged-outcome estimate to −0.033 (s.e. 0.017) and the preferred estimate to −0.002 (s.e. 0.016).

5.2 Robustness

Table 7 reports variations of the preferred specification. Column (1) repeats the preferred estimate. Panel A changes the specification, and Panel B changes the sample, the weighting, and the inference.

Table 7: Robustness of the preferred estimate

Panel A. Specification				
	Preferred	Border-distance bins × year	Shift excludes own district only	Shift uses all districts
	(1)	(2)	(3)	(4)
Cumulative exposure $_{t-1}$	−0.0273***	−0.0399**	−0.0253***	−0.0249***
	(0.0094)	(0.0154)	(0.0077)	(0.0076)
Kleibergen–Paap F	27.40	12.17	36.48	37.17
Observations	6,400	6,400	6,400	6,400
Panel B. Sample, weighting, and inference				
	Trimmed exposure	Population weights	Regional clustering	Long difference
	(5)	(6)	(7)	(8)
Cumulative exposure $_{t-1}$	−0.0455***	−0.0413**	−0.0273***	−0.0245**
	(0.0154)	(0.0190)	(0.0101)	(0.0104)
Kleibergen–Paap F	20.53	12.65	18.25	26.24
Observations	5,740	6,400	6,400	320

Notes: The dependent variable is annual divorces per 1,000 residents of the fixed 1987 population. All columns except (8) use the preferred specification: district fixed effects, state-by-year fixed effects, and baseline controls interacted with year indicators, on 1985–1988 and 1992–2007. Column (2) adds fixed effects for 50-kilometer bins of distance to the former inner-German border interacted with year indicators. Columns (3) and (4) replace the instrument with versions whose shift excludes only the district's own arrivals or includes arrivals in all West German districts (Section 4). Column (5) excludes districts below the 5th and above the 95th percentile of cumulative exposure in 2007. Column (6) weights by 1987 population. Column (7) clusters standard errors by the 81 spatial planning regions (*Raumordnungsregionen*); all other columns cluster by district. Column (8) regresses the change in mean divorce rates between 1985–1987 and 2005–2007 on mean exposure in 2005–2007, instrumented with the mean instrument, with state fixed effects, baseline controls, and the 1987 divorce rate, and reports robust standard errors; the Anderson–Rubin test rejects zero at $p = 0.017$. Anderson–Rubin 95% confidence sets: $[-0.050, -0.010]$ in column (1) and $[-0.091, -0.014]$ in column (2). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column (2) adds fixed effects for 50-kilometer bins of distance to the former inner-German border interacted with year indicators. These absorb shocks that vary with distance to the border, including changes in market access and the phase-out of place-based support for the former *Zonenrandgebiet* between 1991 and 1994, which coincides with the emergence of the divorce decline (Ehrlich & Seidel, 2018; Redding & Sturm, 2008). The estimated decline rises to 0.040. The first stage weakens ($F = 12.2$), but the Anderson–Rubin set still excludes zero. Columns (3) and (4) calculate the aggregate arrival rate from other sample districts and from the full sample, respectively; the esti-

mates remain essentially unchanged (−0.025). Because these variants reweight a largely common aggregate path, they show that the results do not depend on how the shift is constructed, but they are not independent validation.

Panel B shows that the decline is not driven by districts with extreme exposure: excluding districts below the 5th and above the 95th percentile of cumulative exposure in 2007 yields −0.046 (column 5). Weighting districts by their 1987 population yields −0.041, with a weaker first stage (column 6). Clustering standard errors by the 81 spatial planning regions, which allows for correlation among neighboring districts, leaves the estimate statistically significant (column 7). Finally, a long-difference specification that compares mean divorce rates in 2005–2007 with those in 1985–1987 across districts yields −0.025 (column 8).

Two further checks concern the sample. Dropping one state at a time leaves the 2SLS estimate between −0.023 and −0.037 and the change in the share gradient between −0.045 and −0.060 (Appendix Table B.12); the first stage is weak only without Lower Saxony ($F = 8.1$). Excluding the 35 districts of the three Bavarian administrative regions on the Czech border, where Czech commuters were admitted from 1991 (Dustmann et al., 2017), gives −0.030 (s.e. 0.010; $F = 31.4$). Across specifications, the estimated decline ranges from 0.023 to 0.046 per arrival per 1,000 residents. The sign is stable, but the estimates should not be read as a precise structural parameter.

5.3 Pre-reunification trends, a placebo share, and income growth

Under the identifying assumption (13), the coefficients θ_k in equation (11) are zero before reunification. The two long divorce panels described in Section 3.1 allow this to be tested over most of the 1980s; Table 8 reports the tests.

The longer pre-periods show no differential trends[5]. In the districts observed since 1982 and since 1980, the pre-reunification interactions are jointly insignificant ($p = 0.34$ and $p = 0.16$), and the linear pre-trend in the share is close to zero. Over 1985–1988 alone, the interactions are also jointly insignificant, but the linear slope is negative and marginally significant (column 1); the longer windows show that this short-run movement is not part of a longer-run trend. In the same balanced districts, divorce declined after reunification by an average of 0.039 per percentage point of the share in the panel observed since 1982, which is statistically significant, and by a similar 0.033, less precisely estimated, in the smaller panel observed since 1980. Measured against the mean of 1982–1988 instead of 1988 alone, the decline in the panel observed since 1982 is 0.047 (s.e. 0.013). Figure 6 plots the full event study for the districts observed since 1982, and Appendix Figure B.2 for those observed since 1980.

[5]With all district-years observed in 1980–1988, an unbalanced panel in which the set of states changes over the pre-period, the joint test rejects, but the linear pre-trend is close to zero (Appendix B.3).

Table 8: Pre-reunification trends in divorce

	All districts	Balanced since 1982	Balanced since 1980
	(1)	(2)	(3)
Joint pre-trend test (p)	0.261	0.341	0.156
Linear pre-trend	−0.0103*	−0.0012	0.0011
	(0.0053)	(0.0034)	(0.0043)
Mean 1992–2007 coefficient		−0.0390**	−0.0330
		(0.0181)	(0.0244)
Pre-period	1985–88	1982–88	1980–88
Districts	320	241	130
States	8	6	4

Notes: Each column estimates equation (11) for divorces, with the share in percentage points. Balanced panels contain the districts observed in every year from the first pre-period year to 2007. *Joint pre-trend test*: p-value for the hypothesis that all pre-period interactions are zero. *Linear pre-trend*: coefficient on the share multiplied by (year − 1988). Both are estimated on pre-period years only. *Mean 1992–2007 coefficient*: average of the 1992–2007 interactions from the full-window event study on the same districts. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

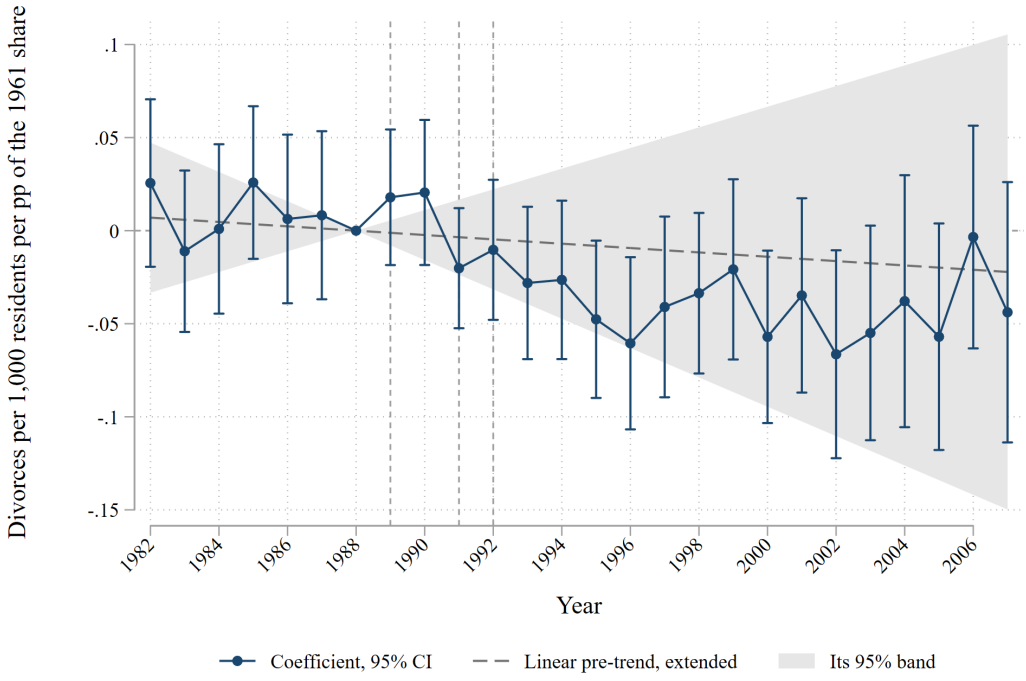


Figure 6: Event study of divorce for districts observed since 1982

Notes: The figure plots the coefficients θ_k of equation (11) for divorces, with the share in percentage points, for the 241 districts in six states observed in every year from 1982 to 2007; standard errors are clustered by district.

Two qualifications apply. First, a pre-period of seven to nine years cannot rule out slow-moving differential trends: a linear trend small enough to be consistent with the

pre-reunification data could, if it continued for two decades, account for much of the post-reunification decline.[6] Measuring the post-reunification coefficients relative to an extrapolated share-specific pre-trend leaves their average negative (−0.030 in the panel observed since 1982) but imprecise (s.e. 0.047). District-specific trends estimated on the post-reunification years are not a better test, because they absorb the response itself (Wolfers, 2006); with them, the estimate is close to zero and imprecise (0.001, s.e. 0.028). Second, in the lagged-outcome specification, districts with higher 1987 divorce rates experienced slower subsequent growth in divorce; allowing for this convergence reduces the lagged-outcome estimate from −0.036 to −0.023, which remains significant. Appendix Table B.6 reports all three trend adjustments.

The share could also capture the settlement of post-war refugees in general rather than ties to the East, and the circumstances of that settlement may have affected later family outcomes. The 1961 census allows a placebo test. Expellees from the former eastern territories who came to the West directly, without first living in the Soviet zone or Berlin, were post-war refugees as well, and the shortage of housing shaped where both they and the East-linked group settled (Burchardi & Hassan, 2013, pp. 1229–1230). Unlike the East-linked group, however, they had not lived in what became East Germany. Burchardi and Hassan (2013, pp. 1249–1250) use them in a falsification test: regions with more direct expellees did not grow faster after reunification, whereas regions with more expellees who had lived in the Soviet zone did. I interact the 1961 share of direct expellees with the cumulative arrival rate of equation (10) and add this placebo instrument to the first stage and the reduced form. Conditional on the instrument, it does not predict arrivals: its coefficient is −0.024 (s.e. 0.033), against 0.566 (s.e. 0.107) for the instrument in the same regression. Per resident of 1961, direct expellees therefore predict at most 7 percent as many arrivals as East-linked residents (the upper end of the 95 percent confidence interval); the arrivals followed ties to East Germany, not the settlement pattern of post-war refugees as such. Nor does the placebo instrument predict divorce (−0.0012, s.e. 0.0017, against −0.0141, s.e. 0.0052). Adding the placebo share interacted with year indicators to the preferred specification leaves the estimate at −0.025 (s.e. 0.009), with a first-stage F of 27.8 (Appendix Table B.9).

The second threat in Section 4.3 includes an income channel: West German regions with more ties to the East grew faster after reunification (Burchardi & Hassan, 2013). Newly assembled district income data show that this channel is absent from the within-state variation that identifies the estimates. I estimate equation (11) with log income per inhabitant from the regional accounts as the outcome and 1989 as the reference year (Appendices A.4 and B.6). By 1995, the end of the period over which Burchardi and

[6] A differential trend of −0.0034 per percentage point of the share per year would generate the average post-reunification coefficient in column (2). It lies inside the confidence interval of the estimated pre-trend, and a test of no pre-trend has about 17 percent power against it (Roth, 2022).

Hassan (2013) measure faster growth, income per inhabitant had not grown faster in districts with stronger ties in any specification (Table B.10, column 2); conditional on 1989 income, the difference is −0.1 percent per percentage point of the share (s.e. 0.2), whereas their estimates imply about 2.4 percent per percentage point of their tie measure (Burchardi & Hassan, 2013, Table III, column 3). The differences in 1992 are small, at most 0.4 percent, and GDP per inhabitant and per person employed did not grow faster either. Consistent with this, the divorce estimate is robust to controlling for income: with year-specific effects of 1989 income it is −0.030 (s.e. 0.011), and with log income per inhabitant as a control −0.027 (s.e. 0.010), against −0.027 (s.e. 0.009) in the preferred specification, and the pre-reunification coefficients remain jointly insignificant ($p = 0.26$; Table B.11). The small remaining income differences could account for the estimated decline only if each percent of income per inhabitant changed the divorce rate by more than 5 percent; for elasticities between −1 and 1, the estimate lies between −0.031 and −0.023 (Conley et al., 2012).

These checks support the identifying assumption: districts with different historical East ties followed similar divorce paths before reunification, and the decline is explained neither by the settlement of post-war refugees as such, nor by shocks that vary with distance to the former border, nor by differences in income growth. The remaining assumption is that no slow-moving differential trend or other early-1990s shock correlated with historical East ties changed divorce.

5.4 Mechanisms

This subsection confronts the evidence with the predictions in Table 1. The marriage results already discriminate between the first and third hypotheses: a thicker partner market should raise both marriages and divorces, whereas higher costs of forming and dissolving households should lower both. Divorces fall and marriages do not rise in any specification, which is inconsistent with Hypothesis 1.

Table 9 uses the breakdown of divorces by filing party, available for Lower Saxony, North Rhine-Westphalia, and Baden-Württemberg, to assess Hypothesis 2. Under that hypothesis, a female-skewed inflow weakens wives' outside options and should reduce wife-initiated divorces more than husband-initiated ones. The estimates use the lagged-outcome specification, whose first stage remains strong in this smaller sample.

In the filing sample, total divorces fall by 0.029 per arrival, similar to the full-sample estimates. Husband-filed divorces do not change, wife-filed divorces fall by 0.011, and jointly filed divorces fall by 0.017, the largest component. The difference between wife- and husband-filed divorces is small and insignificant (−0.005, s.e. 0.009). These results do not support a one-sided shift in bargaining power. The concentration of the point estimates in jointly filed divorces is instead consistent with a change in the joint value

Table 9: Divorces by filing party

	Total	Husband-filed	Wife-filed	Joint	Wife minus husband
	(1)	(2)	(3)	(4)	(5)
Panel A. OLS					
Cumulative exposure $_{t-1}$	−0.0025	−0.0008	−0.0011	−0.0006	
	(0.0049)	(0.0023)	(0.0038)	(0.0029)	
Panel B. 2SLS					
Cumulative exposure $_{t-1}$	−0.0288***	−0.0010	−0.0110	−0.0168*	−0.0049
	(0.0107)	(0.0053)	(0.0083)	(0.0086)	(0.0089)
Kleibergen–Paap F	20.10	20.10	20.10	20.10	20.10
Observations	2,165	2,165	2,165	2,165	2,165
Districts	141	141	141	141	141

Notes: The sample consists of district-years in Lower Saxony, North Rhine-Westphalia, and Baden-Württemberg, 1992–2007, in which divorces are reported by filing party and the three components sum to the total; Lower Saxony does not report the breakdown in 1995 and 1997. Columns (1)–(4) use divorces per 1,000 residents of the fixed 1987 population; the components in columns (2)–(4) sum to column (1). Column (5) uses the change since 1987 in wife-filed minus husband-filed divorces per 1,000 residents. All regressions use the lagged-outcome specification: state-by-year fixed effects, baseline controls, and the 1987 divorce rate. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

of separation, as Hypothesis 3 implies, although none of the components is precisely estimated and the evidence should be read as exploratory.

The magnitude of the decline also argues against Hypothesis 2. Equation (3) implies that one additional young arrival per 1,000 residents lowers the local ratio of men to women aged 20–34 by about 0.0008. The preferred estimate corresponds to a decline in divorce of about 1.1 percent of the mean per arrival, and the lagged-outcome estimate to 1.4 percent. A sex-ratio channel would therefore require divorce to fall by roughly 14 to 18 percent for every one-point reduction in the number of men per 100 women, far beyond the district-level sex-ratio effects estimated for Germany, which are not robustly different from zero (Obersneider et al., 2019). Including the 1989–1990 surge, which was male-dominated, would move local sex ratios in the opposite direction.

Four further results narrow the interpretation. First, births do not rise as they would if every arrival had stayed and had the fertility of West German residents of the same age. Per percentage point of the share, such arrivals alone would raise births by about 0.06 per 1,000 residents (Appendix B.4); instead, births fall by 0.071 (s.e. 0.051; Table 5), with no differential trend before reunification. This is 0.13 below the benchmark (s.e. 0.05), and 0.09 below it (s.e. 0.05) if only arrivals net of departures to the East are counted; a Poisson model gives a fall of 0.8 percent (s.e. 0.5). The benchmark is generous: it attributes all births to residents aged 20–34 and assumes that every arrival stayed, whereas about half of the East German women of the 1965–1974 cohorts who moved to the West after

reunification returned before age 40, and those who returned had elevated first-birth risks afterwards (Kreyenfeld & Vatterrott, 2018). The shortfall is significant at the 5 percent level only if each gross arrival had at least 0.019 births a year, 43 percent of the West German rate; with arrivals counted net of departures to the East, it is not significant even at the full rate ($p = 0.054$). Per arrival, the preferred 2SLS estimate of 0.006 (s.e. 0.027) is also below the benchmark of 0.045, though not significantly, and in the lagged-outcome specification neither total births nor the split between marital and nonmarital births changes significantly. Second, the decline in divorce does not reflect a smaller stock of marriages at risk: it emerges in the mid-1990s, when that stock consisted overwhelmingly of marriages formed before reunification, and marriage formation does not fall robustly. Third, migration between West German districts gives no evidence that arrivals displaced incumbent residents (Borjas, 2006). If young East Germans had led residents to move away, districts would have lost residents to the rest of West Germany as exposure grew. With exposure instrumented as in equation (7), within districts over 1992–2007, one additional young arrival per 1,000 residents changes the annual net migration rate with other West German districts by 0.26 per 1,000 residents (s.e. 0.30) for all ages and by 0.06 (s.e. 0.10) at ages 30–49, with a first-stage F of 29 (Appendix Table B.13). The point estimates have the opposite sign to displacement, but they are imprecise, and because these flows are not observed before 1991, they show only that net migration did not fall as exposure accumulated, not that its level was unaffected. In the cross-section, the mean annual net rate of 1991–2006 is lower by 0.36 per 1,000 residents per percentage point of the share (s.e. 0.31), a difference that cannot be separated from permanent differences between districts.[7]

Fourth, departures to the East lowered the number of residents at risk, but by too little to explain the decline. Historical ties raised departures by three quarters as much as arrivals: per unit of the instrument, cumulative departures of all ages rise by 0.83 per 1,000 residents (s.e. 0.26), against 1.13 (s.e. 0.25) for arrivals, and departures at ages 30–49 by 0.23 (s.e. 0.08). The matrices do not record whether those who left had come from the East earlier. Even if every divorce occurred among residents aged 30–49 and those who left had the average divorce rate of that age group, these departures would lower annual divorces by 0.002 per unit of the instrument, or 14 percent of the reduced-form coefficient of 0.015 in Table 4. Giving every departing resident, whatever the age, the average divorce rate per resident yields the same share.[8]

[7]Per unit of the annual change in the instrument, which is the share multiplied by the annual arrival rate in the other states, annual young arrivals from the East rise by 0.42 per 1,000 residents (s.e. 0.07) and annual young arrivals from other West German districts by 1.17 (s.e. 0.25), in regressions on 1991–2006 with district and state-by-year fixed effects. Arrivals from other West German districts include East Germans who moved on within the West, in which case measured exposure understates the inflow that ties attracted; they may also show that districts with stronger ties attracted more movers of any origin in years of high East–West migration.

[8]The calculation divides mean annual divorces per 1,000 residents in 1992–2007, 2.52, by the number

Taken together, the evidence rejects Hypothesis 1 and does not support Hypothesis 2. The pattern of fewer divorces without more marriages has the signs that higher costs of forming and dissolving households predict, but the district data cannot test this channel directly: the available housing measures describe the housing stock in 1987, not rent growth after 1989, and the decline does not vary with them (Section 5.5). Births rise less than a generous fertility benchmark implies, but whether the shortfall is significant depends on how many arrivals stayed and on their fertility, so the evidence cannot establish whether the decline in divorce is specific to marital dissolution or part of a broader change in family events in districts with strong ties. Of the non-behavioral mechanisms, departures to the East account for about a seventh of the decline, and displacement to other West German districts is not detected, although the district data cannot exclude it.

5.5 Heterogeneity

I next examine whether the response varies across local environments, by district type and by distance to the former inner-German border. Because the preferred specification has limited first-stage power in subsamples, the heterogeneity tables report OLS and 2SLS estimates with state-by-year fixed effects only and in the lagged-outcome specification. These estimates are descriptive, and whenever the Kleibergen–Paap statistic is low, the corresponding 2SLS coefficient is uninformative.

Table 10 distinguishes independent cities from counties. In counties, the lagged-outcome estimate is -0.019 with a strong first stage ($F = 37.7$). In independent cities, the estimate is close to zero, but the first stage is weak ($F = 11.2$), so the estimate is uninformative rather than evidence of no effect.

Table 11 splits districts by distance to the former inner-German border. Within 100 kilometers, the lagged-outcome estimate is -0.028 with an adequate first stage ($F = 17.1$). Farther from the border, the first stage is weak ($F = 3.7$ for 100–200 kilometers and 4.7 beyond 200 kilometers), and the estimates are uninformative. In both tables, the specifications with state-by-year fixed effects only again yield positive or imprecise coefficients, reflecting the persistent level differences discussed in Section 5.1. In the lagged-outcome specification, the 2SLS response is thus precisely estimated only in counties and near the former border, because the first stage is weak elsewhere.[9]

of residents aged 30–49 per 1,000 residents in the 1987 census, 275. The cohorts of the baby boom entered this age group after 1987, so the 1987 figure understates the number of residents aged 30–49 in 1992–2007 and overstates their divorce rate.

[9] In the within-district specification with district and state-by-year fixed effects on 1992–2007 (as in column 1 of Appendix Table B.7), which avoids estimating nine cross-sectional control coefficients within each distance band, the first stage remains informative farther from the border: the 2SLS estimate is -0.0381^{***} (s.e. 0.0090, $F = 13.81$) within 100 kilometers, -0.1107^{***} (s.e. 0.0302, $F = 15.94$) at 100–200 kilometers, and -0.0613 (s.e. 0.0378, $F = 10.82$) beyond 200 kilometers; in counties it is -0.0360^{***} (s.e. 0.0117, $F = 31.04$) and in independent cities 0.0242 (s.e. 0.0257, $F = 8.27$).

Table 10: Heterogeneity by district type

	OLS		2SLS	
	(1)	(2)	(3)	(4)
Panel A. Independent cities (kreisfreie Städte)				
Cumulative exposure $_{t-1}$	−0.0064	−0.0076	−0.0015	−0.0054
	(0.0054)	(0.0048)	(0.0237)	(0.0187)
Kleibergen–Paap F			6.74	11.24
Observations	1,376	1,376	1,376	1,376
Panel B. Counties (Landkreise)				
Cumulative exposure $_{t-1}$	0.0061	0.0029	0.0437***	−0.0185**
	(0.0043)	(0.0035)	(0.0142)	(0.0078)
Kleibergen–Paap F			26.19	37.65
Observations	3,744	3,744	3,744	3,744
State × year fixed effects	✓	✓	✓	✓
Baseline controls		✓		✓
1987 divorce rate		✓		✓

Notes: The dependent variable is annual divorces per 1,000 residents of the fixed 1987 population, 1992–2007. Columns (3) and (4) instrument lagged cumulative exposure with the shift-share instrument. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The reduced form does not depend on the first stage. Table 12 compares the change in the share gradient, θ in equation (12), across groups of districts, adding group-by-year effects. The decline is present in independent cities (−0.038, s.e. 0.014) and in counties (−0.065, s.e. 0.014), and the difference between them is not significant ($p = 0.17$). It is of similar size within 100 kilometers of the former border (−0.051, s.e. 0.012) and beyond (−0.060, s.e. 0.014). It also does not vary with the state of the housing market in 1987: the gradient is −0.057 where the number of households per dwelling was above the state median and −0.047 where it was below, and −0.044 against −0.055 for districts above and below the median number of dwellings per residential building. The decline is therefore not confined to one type of district. These measures describe the 1987 housing stock rather than rent growth after 1989, so the comparison is a weak test of Hypothesis 3

6 Conclusion

This paper studies how young East–West migration after German reunification affected marriage formation and marital dissolution in receiving West German districts. I combine a newly assembled panel of marriage, divorce and birth outcomes for 320 districts over 1985–2007 with observed cumulative arrivals aged 18–29 from 1991 onward. To address endogenous destination choice, I instrument exposure using 1961 East-linked settlement

Table 11: Heterogeneity by distance to the former inner-German border

	OLS		2SLS	
	(1)	(2)	(3)	(4)
Panel A. Within 100 km				
Cumulative exposure $_{t-1}$	0.0018	−0.0054	0.0080	−0.0281***
	(0.0045)	(0.0041)	(0.0090)	(0.0082)
Kleibergen–Paap F			8.81	17.09
Observations	1,360	1,360	1,360	1,360
Panel B. 100–200 km				
Cumulative exposure $_{t-1}$	0.0047	0.0023	0.0592*	−0.2059*
	(0.0064)	(0.0063)	(0.0351)	(0.1141)
Kleibergen–Paap F			9.27	3.71
Observations	2,000	2,000	2,000	2,000
Panel C. Above 200 km				
Cumulative exposure $_{t-1}$	0.0120	0.0024	0.1109**	−0.0041
	(0.0081)	(0.0073)	(0.0508)	(0.0367)
Kleibergen–Paap F			8.45	4.69
Observations	1,744	1,744	1,744	1,744
State × year fixed effects	✓	✓	✓	✓
Baseline controls		✓		✓
1987 divorce rate		✓		✓

Notes: The dependent variable is annual divorces per 1,000 residents of the fixed 1987 population, 1992–2007. Distance is measured from each district to the former inner-German border. Columns (3) and (4) instrument lagged cumulative exposure with the shift-share instrument. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

shares interacted with a cumulative young-arrival rate constructed outside each district's own state. The exposure measure includes arrivals of both sexes, and the estimates do not identify a female-specific effect.

The main empirical result is that divorce declined after reunification in districts with stronger historical ties to the East, by 2 percent for each percentage point of the 1961 share relative to 1985–1988. In the preferred specification, which includes district and state-by-year fixed effects and year-specific effects of predetermined district characteristics, one additional cumulative young arrival per 1,000 residents of the fixed 1988 population corresponds to 0.027 fewer annual divorces per 1,000 residents of the fixed 1987 population. A one-standard-deviation increase in exposure corresponds to approximately 9 percent of the mean divorce outcome. The first stage is strong, and the weak-instrument-robust confidence set excludes zero. The negative estimate persists across alternative instrument constructions, border-distance controls, income controls, sample restrictions, the exclusion of any single state, population weighting and regional clustering.

The evidence for marriage formation is less conclusive. The preferred marriage es-

Table 12: Reduced-form gradient by type of district

Group indicator G_i	County	Within 100 km of the border	Households per dwelling above state median	Dwellings per building above state median
	(1)	(2)	(3)	(4)
1961 share × post, $G_i = 0$	−0.0376***	−0.0602***	−0.0467***	−0.0549***
	(0.0144)	(0.0143)	(0.0147)	(0.0159)
1961 share × post, $G_i = 1$	−0.0649***	−0.0512***	−0.0574***	−0.0438***
	(0.0144)	(0.0118)	(0.0103)	(0.0112)
Difference	−0.0273	0.0090	−0.0108	0.0112
	(0.0197)	(0.0145)	(0.0129)	(0.0160)
Observations	6,400	6,400	6,400	6,400

Notes: Divorces per 1,000 residents of 1987; 320 districts, 1985–1988 and 1992–2007. Each column adds to the preferred specification the interaction of the share-by-post term with the group indicator and group-by-year fixed effects, so the difference compares share gradients and not group trends. $G_i = 0$ denotes independent cities in column (1), districts more than 100 kilometers from the former inner-German border in column (2), and districts at or below the state median in columns (3) and (4); housing variables are from the 1987 census. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

timate is negative, at −0.021, but statistically imprecise, and the estimated decline is sensitive to the specification and the inclusion of districts that attract weddings from elsewhere. The results therefore do not establish a robust decline in marriage formation. Births do not rise as they would if every arrival had stayed and had the fertility of West German residents. In the share reduced form they fall by 0.7 percent of the mean per percentage point, though not significantly, and they fall significantly short of the arrivals' own fertility only under generous assumptions about how many arrivals stayed and how many children they had. The district data therefore cannot establish whether the decline in divorce is specific to marital dissolution or part of a broader change in family events in districts with strong historical ties.

The timing of the divorce response also shapes its interpretation. The decline emerges in the mid-1990s, a few years after the largest wave of 1989–1990, and does not deepen in proportion to the arrivals measured from 1991 onward. Measured exposure also counts gross arrivals, whereas historical ties raised departures to the East by three quarters as much as arrivals. This pattern limits its interpretation as a constant marginal effect of each additional arrival. Moreover, district-level exposure begins in 1991 and omits the substantial inflows of 1989 and 1990. With the arrival rate extended to these years using state totals, cumulative predicted arrivals since 1989 describe the path of the decline better than a one-time shift, consistent with these earlier arrivals contributing to the estimated response.

The heterogeneity results remain descriptive. Under the lagged-outcome specification, instrumental-variable estimates are precise only in counties and in districts within 100

kilometers of the former inner-German border, where the first stage is strong. In the reduced form, which does not depend on the first stage, the decline is present in independent cities and in counties, near and far from the former border, and in districts with tight and with slack housing markets before reunification. These comparisons describe the geographical pattern of the response but do not identify its mechanism.

The mechanism evidence is similarly limited. The filing decomposition does not establish a statistically significant difference between wife- and husband-initiated divorce responses. Jointly filed divorces have the largest negative point estimate, although the component estimates are imprecise. The negative aggregate divorce response is difficult to reconcile with a simple account in which expanded partner availability increases marital dissolution, but the evidence does not isolate a sex-ratio or bargaining channel. The pattern of fewer divorces without more marriages has the signs that higher costs of forming and dissolving households predict, but the district data cannot test this channel directly: the available housing measures describe the housing stock in 1987, not rent growth after 1989. Net migration with other West German districts did not fall as arrivals accumulated, and departures to the East account for about a seventh of the decline. Because the outcomes are event counts scaled by fixed baseline populations, the estimates still cannot distinguish changes in couples' behavior from changes in the composition of the couples residing in a district.

The main caveat remains identification. Balanced district panels extending back to 1980 and 1982 do not reject the null of no differential pre-reunification divorce trends. However, these tests have limited power against gradual divergence, and estimates measured relative to an extrapolated pre-reunification trend remain negative but imprecise. Border-distance-by-year fixed effects absorb shocks that vary with distance to the former border, a placebo share of expellees who had not lived in the Soviet zone predicts neither arrivals nor divorce, and newly assembled district income data show that districts with stronger ties did not experience faster income growth after reunification; the estimate is robust to controlling for income. A causal interpretation therefore requires that, conditional on the fixed effects and predetermined characteristics, historical settlement ties affected subsequent family outcomes through migration exposure rather than through other channels.

The paper contributes a harmonized historical district panel and evidence linking reunification-era migration exposure to lower divorce activity in receiving West German districts. Under the identifying assumptions, the results indicate that the consequences of young internal migration extended to marital dissolution. Establishing whether this reflected greater stability within existing marriages, changes in the resident population, or adjustments in local economic conditions requires evidence beyond aggregate district counts.

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A Data Construction

A.1 Divorce counts

The replication files list every corrected cell of the divorce panel, with the original transcription, the corrected count, the type of error and the source table. In a few state-years, the district counts could be compared only with the totals printed for administrative regions or for the state: Baden-Württemberg in 1981, 1982, 1984 and 2001, Lower Saxony in 2000 and 2001, and the Saarland in 1980–1984.

In four pairs of a city and its surrounding county, the published counts appear to assign divorces to the wrong member of the pair: Kassel in 2005–2007, where the Hessian statistical office notes the error, and Aachen, Oldenburg and Osnabrück in 2006–2007. These 18 district-years enter the analysis as published; dropping them changes the preferred estimate from -0.027 to -0.025 (s.e. 0.009).

The counts for the early 1980s contain abrupt changes in level. Of the 1,721 year-on-year changes observed for the 241 districts of the panel balanced since 1982, 73 exceed 40 percent in absolute value, 70 of them increases. Of these 73, 42 fall in five state-years: Bavaria in 1983 (13 districts), 1984 (6) and 1987 (10), North Rhine-Westphalia in 1981 (7), and Hesse in 1984 (6). The state-by-year fixed effects absorb the part of these changes that is common to the districts of a state. In 1985–1988, the counts used for the long pre-periods are identical to those of the main panel in all 1,280 district-years.

A.2 Migration counts

Apart from the merger that formed the Region Hannover in 2001, the West German districts do not change between 1991 and 2007, so the migration matrices need no boundary conversion for the 320 sample districts. Recomputing the 1991 state totals of young arrivals, and the 2001 total for Baden-Württemberg, directly from the original matrix files reproduces the processed counts exactly. For departures to the East and for moves between West German districts, I read the same matrices a second time, by district of origin as well as of destination. Cumulative young arrivals rebuilt in this second pass equal the exposure variable in every district and year. Figure 2 plots these counts, summed over

the 324 West German districts outside the city-states.

The arrival rate $g_{-s,\tau}$ of equation (10) requires only state totals and can therefore be extended to 1989 and 1990, for which no district matrices exist. For these years, the annual volumes *Gebiet und Bevölkerung* of the Federal Statistical Office report arrivals of Germans from the GDR by receiving state, with East Berlin listed separately and excluded here, in total and by sex but not by age; I digitized them from the printed volumes (Statistisches Bundesamt, 1987–2001). I scale each state's total by the ratio, in 1991, of arrivals aged 18–29 in its sample districts, taken from the matrices, to the state total reported for 1991. With arrivals of all ages in the numerator, this 1991 ratio is between 1.01 and 1.03 in the seven states whose districts are all in the sample and 0.82 in Lower Saxony, whose total includes the four districts outside the sample. The scaled totals imply 141,000 young arrivals in the sample districts in 1989 and 155,000 in 1990, against 97,000 in 1991. The scaling assumes that the arrivals of 1989–1990 had the age composition of 1991; because the early wave included more families (Section 2), it more likely overstates their number of young adults, so all results are also reported with arrivals of all ages, which require no such assumption. Applied to 1991, the same construction reproduces the arrival rate computed from the matrices exactly.

A.3 The 1961 East-linked share

The census district file reports R_i and $P_{i,1961}$ for the 565 districts of 1961, which the territorial reforms of the late 1960s and the 1970s merged and divided. I assign the counts of each 1961 district to the 1991 district or districts that contain its territory. R_i and $P_{i,1961}$ are converted in the same way, and s_i is computed from the converted counts. The census district file also counts expellees from the former eastern territories, in total and for those who came to the Federal Republic through the Soviet zone or Berlin. I convert both counts in the same way; their difference, divided by $P_{i,1961}$, is the share s_i^P of expellees who came directly, the placebo of Section 5.3.

Table A.1 sums the converted counts by state and compares them with the published census results. In the seven states whose districts are all in the sample, the sums equal the published counts up to the rounding of the published table, and the shares agree to the first decimal. Because any conversion that allocates every 1961 district in full preserves state totals, the comparison verifies the completeness of the census file and of the conversion, not the allocation of divided districts within a state. The Lower Saxony sums cover only the 43 sample districts and are therefore not comparable with the published total.

Table A.1: The 1961 East-linked population by state

	Published		Sample districts		
	Thousands	Share (%)	Number	Thousands	Share (%)
Schleswig-Holstein	114	4.9	15	114.3	4.9
Lower Saxony	356	5.4	43	252.0	4.7
North Rhine-Westphalia	909	5.7	54	908.8	5.7
Hesse	302	6.3	26	301.7	6.3
Rhineland-Palatinate	128	3.7	36	127.6	3.7
Baden-Württemberg	416	5.4	44	415.9	5.4
Bavaria	294	3.1	96	294.2	3.1
Saarland	23	2.1	6	22.8	2.1

Notes: Residents in 1961 who had lived in the Soviet zone or East Berlin in 1939 and moved to the Federal Republic after the war, with their children born since. Published: state results of the 1961 census reported by Ritter & Niehuss (1987, p. 31). Sample districts: sums of the converted counts R_i and $P_{i,1961}$ over the sample districts of each state. The Lower Saxony sample excludes the city and the district of Hannover, Göttingen and Osterode am Harz.

A.4 District income

District income comes from the regional accounts of the statistical offices. For 1989 and 1992, I digitized primary and disposable income of private households, as totals and per inhabitant, for all districts of the former West Germany from Arbeitskreis Volkswirtschaftliche Gesamtrechnungen der Länder (VGRdL, 1997), which reports both years in the vintage of autumn 1995. For 1995–2007, the same aggregates come from the August 2008 vintage (VGRdL, 2009b), and GDP and employment at the place of work for 1992 and 1994–2007 from its companion volume (VGRdL, 2009a). District income is not reported for 1990, 1991, 1993 and 1994. Deutsche Mark values are converted at 1.95583 per euro.

The digitized values pass three accounting checks: district totals add up to the printed totals of administrative regions and states; total and per-inhabitant income imply the same population in the tables for primary and disposable income, up to rounding; and the implied 1992 population matches that of the 2008 vintage within 0.004 percent in every district. One misread digit, in the 1989 total disposable income of Ulm, was corrected.

The two vintages follow different revisions of the national accounts. Between 1992 and 1995, the revision raised income per inhabitant in poorer rural districts relative to the richest ones: the dispersion of these changes across districts (a standard deviation of 0.057 log points) exceeds that of three-year changes within either vintage (0.040 and 0.029). Because districts with stronger historical ties had higher incomes in 1989—within states, one percentage point of the share goes with about 4.5 percent higher disposable income per inhabitant (s.e. 0.5)—comparisons across the two vintages are also reported with year-specific effects of log 1989 income, which absorb any revision that is linear in log income. Comparisons within a vintage need no such adjustment.

B Additional Results

B.1 Dose-response and timing

Table B.1 reports the comparisons across quintiles of the 1961 share, Table B.2 the effect of exposure on divorce by period in the lagged-outcome specification, and Table B.3 the timing of the change in the preferred specification (Section 5.1).

Table B.4 compares descriptions of that timing. The fits use the sixteen coefficients $\hat{\theta} = (\hat{\theta}_{1992}, \ldots, \hat{\theta}_{2007})'$ of equation (11), estimated in the preferred specification with 1985–1988 as the reference period, so that their mean equals $\hat{\theta}$ in Table 5, column (1). Each description restricts $\hat{\theta} = G\beta + e$, and

$$\hat{\beta} = \left(G'\hat{V}^{-1}G\right)^{-1}G'\hat{V}^{-1}\hat{\theta}, \qquad J = \left(\hat{\theta} - G\hat{\beta}\right)'\hat{V}^{-1}\left(\hat{\theta} - G\hat{\beta}\right), \tag{14}$$

where $\hat{V}$ is the cluster-robust covariance matrix of $\hat{\theta}$ and the columns of G are the regressors named in the table. If the restriction holds, J is χ^2 with $16 - \dim\beta$ degrees of freedom, and the difference in J between nested restrictions is $\chi^2(1)$.

Panel A shows that counting the arrivals of 1989–1990 changes the description of the timing. Cumulative arrivals since 1991, the measure in the instrument, fit worse than a shift combined with them. Cumulative arrivals since 1989, with a single parameter, fit as well as that two-parameter combination and are not rejected at the 5 percent level, whether the 1989–1990 totals are scaled to young adults or arrivals of all ages are counted throughout. Panel B examines whether divorce instead responds to recent arrivals, measured by the mean arrival rate over the three years before t. With the 1989–1990 arrivals included, the recent inflow does not improve on a one-time shift in the expected direction: its coefficient is positive. Restricted to 1994–2007, whose windows lie within the years covered by the matrices, it has the expected sign, but the improvement in fit has $p = 0.053$. The data therefore do not single out a response to recent arrivals.

B.2 Event-study coefficients

Table B.5 reports the coefficients of equation (11) for divorces, which Figure 5 plots, and for marriages.

B.3 Pre-reunification trends

Figure B.2 shows the event study for the districts observed since 1980. Table B.6 reports the test on all district-years observed before reunification and the trend adjustments discussed in Section 5.3.

Table B.1: Change after reunification by quintile of the 1961 share

	Divorces (1)	Marriages (2)
Second quintile × post	−0.0911**	0.1485
	(0.0432)	(0.1385)
Third quintile × post	−0.2449***	−0.0273
	(0.0600)	(0.1474)
Fourth quintile × post	−0.3160***	−0.1559
	(0.0650)	(0.1767)
Top quintile × post	−0.3532***	−0.1231
	(0.0683)	(0.1921)
Joint test, $F(4, 319)$ [p-value]	7.88 [<0.001]	1.03 [0.391]
Observations	6,400	6,400

Notes: Reduced-form regressions of the outcome on indicators for quintiles of the 1961 East-linked share, defined over all 320 districts, interacted with an indicator for 1992–2007; the bottom quintile is the reference. Fixed effects, controls and sample are those of the preferred specification: district fixed effects, state-by-year fixed effects, and baseline controls interacted with year indicators, on 1985–1988 and 1992–2007. Outcomes per 1,000 residents of 1987. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2: Effect of exposure on divorces by period

	1992–1996 (1)	1997–2001 (2)	2002–2007 (3)
Cumulative exposure $_{t-1}$	−0.0684**	−0.0163	−0.0212**
	(0.0316)	(0.0111)	(0.0085)
Kleibergen–Paap F	15.78	23.22	28.01
Observations	1,600	1,600	1,920

Notes: 2SLS estimates in the lagged-outcome specification (state-by-year fixed effects, baseline controls, and the 1987 divorce rate), estimated separately for each period. Divorces per 1,000 residents of 1987. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.4 Births and alternative within-district specifications

Arrivals raise local birth counts at least through their own fertility, which gives a benchmark for the birth estimates. Attributing all births to residents aged 20–34, the sample means in Table 2 imply about $10.73/237 \approx 0.045$ births per resident of that age per year, where 237 is the number of residents aged 20–34 per 1,000 residents in 1987. If every measured arrival stayed and had this fertility, births per 1,000 residents would rise by about 0.045 per unit of exposure; return and onward moves, and the ageing of earlier arrivals, lower this benchmark further. An estimate far above it would indicate that the instrument picks up differences between districts other than the arrivals. In the share

Table B.3: Timing of the change in divorce by historical East-linked share

	1992–1996 (1)	1997–2001 (2)	2002–2007 (3)
Panel A. Reduced form by period, one regression			
1961 share × period	−0.0453***	−0.0378***	−0.0745***
	(0.0103)	(0.0117)	(0.0154)
Equality of the three coefficients (p)		0.023	
Observations		6,400	
Panel B. 2SLS, each period with 1985–1988			
Cumulative exposure$_{t-1}$	−0.1179***	−0.0351**	−0.0338***
	(0.0411)	(0.0143)	(0.0103)
Kleibergen–Paap F	15.83	21.70	26.94
Observations	2,880	2,880	3,200
Panel C. Shift against dose, one regression			
1961 share × post		−0.0378***	
		(0.0126)	
Instrument Z_{it}		−0.0078	
		(0.0062)	
Observations		6,400	

Notes: Divorces per 1,000 residents of 1987; preferred specification (district fixed effects, state-by-year fixed effects, baseline controls interacted with year indicators), 320 districts. Panel A interacts the 1961 East-linked share, in percentage points, with indicators for three periods; the omitted period is 1985–1988. Panel B estimates equation (7) on 1985–1988 and the period of the column. Panel C enters the share interacted with an indicator for 1992–2007 together with the instrument of equation (10), which is the share multiplied by the cumulative arrival rate. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

reduced form of equation (12), the benchmark is 0.045 times the increase in cumulative young exposure per percentage point of the share after reunification, 1.32 (s.e. 0.24), or 0.060 births per 1,000 residents. Counting only arrivals net of departures to the East, whose increase is 0.48 (s.e. 0.12), it is 0.022. Regressing births minus 0.045 times exposure on the share gives the distance to the benchmark with a standard error that accounts for the estimated increase in exposure. Repeating this regression with fertility between zero and 0.045 gives the lowest fertility at which the distance is significant at the 5 percent level: 0.019 births per gross arrival per year, 43 percent of the benchmark rate. Counting net arrivals, the distance is not significant at any fertility up to 0.045 ($p = 0.054$ at 0.045). Section 5.4 reports both.

Table B.7 uses this benchmark to assess the specification. With district and state-by-year fixed effects alone (column 1), births rise by 0.178 per unit of exposure, four times the benchmark, and divorces fall by 0.054, twice the preferred estimate. Letting the baseline controls have year-specific effects (column 2) lowers the birth estimate to 0.056, and adding the pre-reunification years (column 3) lowers it to 0.006; neither dif-

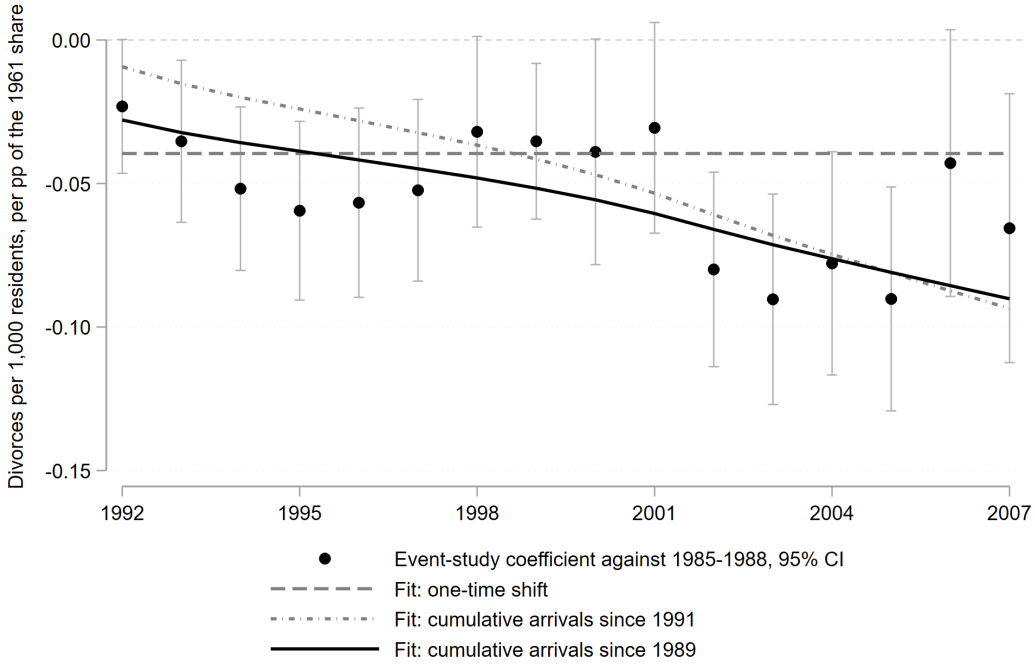


Figure B.1: Divorce by historical East-linked share: event-study coefficients and fitted paths

Notes: Points: coefficients $\hat{\theta}_k$ of equation (11) with 1985–1988 as the reference period, with 95% confidence intervals clustered by district. Lines: minimum-distance fits of equation (14) from Panel A of Table B.4; cumulative arrivals since 1989 use state totals for 1989–1990 (Appendix A.2).

fers significantly from the benchmark. The year-specific effects of the baseline controls therefore absorb differential trends that the share would otherwise pick up, which is why the preferred specification includes them and does not use the larger divorce estimate of column (1). Table B.8 reports births by the parents' marital status.

B.5 Placebo share

Table B.9 reports the estimates behind the placebo test in Section 5.3. The placebo instrument replaces the East-linked share in equation (10) with the 1961 share of expellees who came to the Federal Republic directly, s_i^P (Appendix A.3), and keeps the arrival rate:

$$Z_{it}^P = s_i^P \sum_{\tau=1991}^{t-1} g_{-s(i),\tau}. \tag{15}$$

Because both instruments use the same arrival rate and both shares are fractions of the 1961 population, the ratio of their coefficients in columns (1) and (2) compares the two groups per resident of 1961.

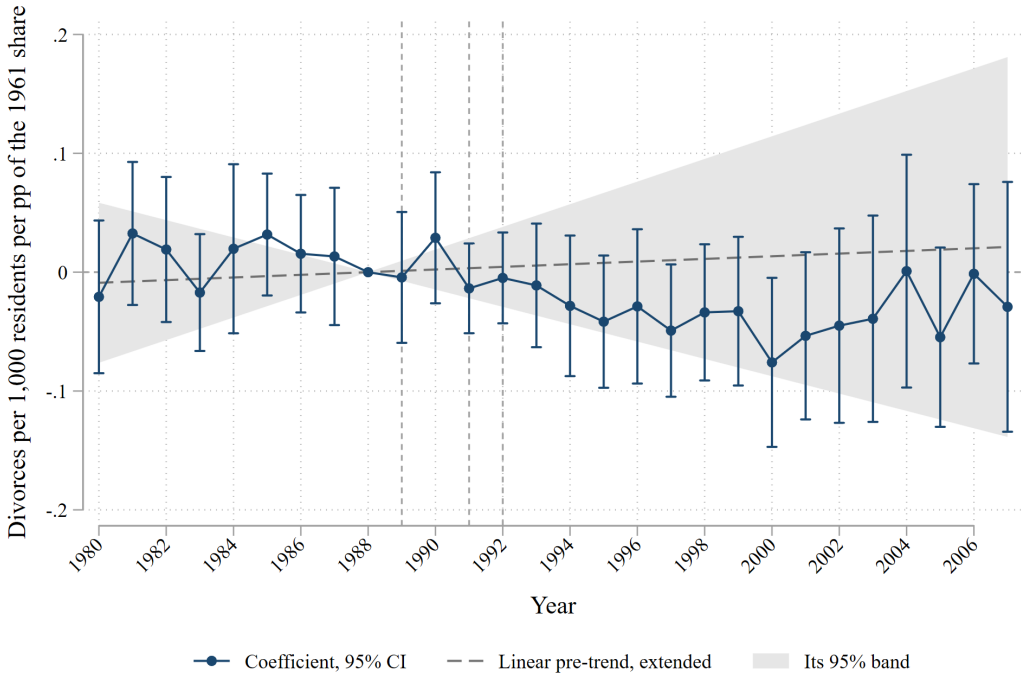


Figure B.2: Event study of divorce for districts observed since 1980

Notes: As Figure 6, for the 130 districts in North Rhine-Westphalia, Hesse, Baden-Württemberg and the Saarland observed in every year from 1980 to 2007. The dashed line extends the linear pre-trend estimated on 1980–1988, and the shaded area is its 95% confidence band.

Table B.4: Timing of the divorce decline and the arrivals of 1989–1990

	Shift	Cumulative arrivals	Recent inflow	J (df)	p
Panel A. One-time shift and cumulative arrivals					
One-time shift	−0.0395			36.58 (15)	0.001
	(0.0083)				
Since 1991		−0.0233		28.66 (15)	0.018
		(0.0042)			
Shift + since 1991	−0.0202	−0.0176		24.59 (14)	0.039
	(0.0100)	(0.0051)			
Since 1989, ages 18–29		−0.0173		24.59 (15)	0.056
		(0.0029)			
Since 1989, all ages		−0.0074		24.42 (15)	0.058
		(0.0013)			
Panel B. Recent inflow: arrivals in the three preceding years					
Shift + inflow, ages 18–29	−0.0618		0.0643	33.04 (14)	0.003
	(0.0144)		(0.0341)		
Shift + inflow, all ages	−0.0679		0.0354	32.22 (14)	0.004
	(0.0159)		(0.0170)		
Since 1989 + inflow, ages 18–29		−0.0176	0.0044	24.55 (14)	0.039
		(0.0034)	(0.0228)		
Since 1989 + inflow, all ages		−0.0074	−0.0007	24.41 (14)	0.041
		(0.0014)	(0.0102)		
Shift + inflow, 1994–2007	−0.0115		−0.1749	24.52 (12)	0.017
	(0.0230)		(0.0906)		

Panel C. Recent inflow added to the path named in the row

	$\chi^2(1)$	p
Shift, ages 18–29	3.55	0.060
Shift, all ages	4.37	0.037
Since 1989, ages 18–29	0.04	0.847
Since 1989, all ages	0.01	0.941
Shift, 1994–2007	3.73	0.053

Notes: Minimum-distance fits of equation (14) to the coefficients $\hat{\theta}_k$ of equation (11) for divorces, 1992–2007 (1994–2007 in the rows so marked), estimated in the preferred specification with 1985–1988 as the reference period; standard errors in parentheses. *Shift*: a constant. *Cumulative arrivals*: the arrival rate $g_{-s,\tau}$ of equation (10) summed through $k-1$, from 1991 as in the instrument or from 1989, with 1989–1990 from state totals (Appendix A.2). *Recent inflow*: the mean of $g_{-s,\tau}$ over $k-3,\ldots,k-1$, including 1989–1990. *Ages 18–29*: state totals for 1989–1990 scaled by the 1991 ratio of young arrivals in the sample districts to the state total; *all ages*: arrivals of all ages in every year. Regressors are multiplied by 0.01 and averaged over the eight leave-state-out series with weights equal to each state's within-state sum of squared deviations of the share, so coefficients are in the units of the coefficient on Z_{it}; all-age coefficients are smaller because arrivals of all ages are about twice as numerous as young arrivals. Panel C: difference in J between the path named in the row and the same path with the recent inflow added; in the rows for the shift with 1989–1990 included, the inflow enters with a positive coefficient (Panel B).

Table B.5: Event-study coefficients for divorces and marriages

	Divorces		Marriages	
Year	Coefficient	Std. error	Coefficient	Std. error
1985	0.0286*	(0.0158)	−0.0066	(0.0231)
1986	0.0267	(0.0162)	0.0144	(0.0236)
1987	0.0092	(0.0148)	0.0105	(0.0170)
1988	Reference year			
1989	0.0068	(0.0138)	−0.0135	(0.0220)
1990	0.0124	(0.0122)	−0.0004	(0.0217)
1991	−0.0003	(0.0121)	0.0008	(0.0227)
1992	−0.0070	(0.0140)	−0.0236	(0.0258)
1993	−0.0192	(0.0152)	−0.0427	(0.0345)
1994	−0.0357**	(0.0150)	−0.0605*	(0.0318)
1995	−0.0434***	(0.0154)	−0.0716**	(0.0286)
1996	−0.0406**	(0.0165)	−0.0780**	(0.0344)
1997	−0.0362**	(0.0164)	−0.0805**	(0.0378)
1998	−0.0159	(0.0177)	−0.0853**	(0.0364)
1999	−0.0192	(0.0155)	−0.0631	(0.0386)
2000	−0.0229	(0.0203)	−0.0556	(0.0368)
2001	−0.0145	(0.0212)	−0.0847**	(0.0378)
2002	−0.0638***	(0.0197)	−0.0637*	(0.0375)
2003	−0.0742***	(0.0212)	−0.0827**	(0.0417)
2004	−0.0617***	(0.0221)	−0.0527	(0.0425)
2005	−0.0741***	(0.0222)	−0.0449	(0.0420)
2006	−0.0268	(0.0254)	−0.0388	(0.0434)
2007	−0.0495*	(0.0261)	−0.0656	(0.0417)
Joint test, 1985–1987: $F(3, 319)$	1.34 [$p = 0.261$]		0.75 [$p = 0.525$]	
Observations	7,360		7,360	

Notes: Coefficients θ_k of equation (11), with the share in percentage points: district fixed effects, state-by-year fixed effects, and baseline controls interacted with year indicators; 320 districts, 1985–2007, including the transition years 1989–1991. Outcomes per 1,000 residents of 1987. Standard errors clustered by district. For divorces, the joint test comes from the same specification estimated on 1985–1988, which yields identical coefficients for 1985–1987. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.6: Unbalanced pre-period test and trend adjustments, divorce

	Estimate	KP F	Districts	Observations
Panel A. All district-years observed in 1980–1988				
Joint pre-trend test (p)	0.021		320	2,278
Linear pre-trend	−0.0020		320	2,278
	(0.0029)			
Panel B. Mean 1992–2007 coefficient, net of the share-specific pre-trend				
Balanced since 1982	−0.0300		241	6,266
	(0.0465)			
Balanced since 1980	−0.0608		130	3,640
	(0.0591)			
Panel C. 2SLS with trend controls, 1992–2007				
District-specific trends	0.0013	49.88	320	5,120
	(0.0284)			
Lagged outcome with convergence	−0.0226***	27.10	320	5,120
	(0.0086)			
1987 divorce rate × years since 1991	−0.0250***			
	(0.0042)			
1985–1988 change × years since 1991	−0.0070			
	(0.0084)			

Notes: Divorces per 1,000 residents of 1987. KP F: Kleibergen–Paap first-stage F statistic. Panel A repeats the pre-period tests of Table 8 on all district-years observed in 1980–1988, an unbalanced panel in which states enter in different years (Section 3.1). Panel B replaces the pre-reunification share-by-year terms of the full-window event study with a linear trend in the share and averages the 1992–2007 coefficients, which then measure deviations from the extrapolated trend; Table 8 reports the unadjusted averages. In Panel C, the first row uses district fixed effects, district-specific linear trends and state-by-year fixed effects. The second row adds to the lagged-outcome specification the 1987 divorce rate and the change in the mean divorce rate from 1985–1986 to 1987–1988, each interacted with the number of years since 1991; both are measured per 1,000 residents. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.7: Alternative within-district specifications

	District and state-year FE (1)	+ Baseline controls × year (2)	Preferred (3)
Divorces	−0.0537***	−0.0152	−0.0273***
	(0.0090)	(0.0103)	(0.0094)
Marriages	−0.0447**	0.0049	−0.0213
	(0.0173)	(0.0186)	(0.0195)
Live births	0.1780***	0.0556***	0.0060
	(0.0269)	(0.0195)	(0.0274)
Kleibergen–Paap F	57.69	29.01	27.40
Observations	5,120	5,120	6,400
District fixed effects	✓	✓	✓
State × year fixed effects	✓	✓	✓
Baseline controls × year		✓	✓
Pre-reunification years (1985–1988)			✓

Notes: 2SLS estimates of the effect of lagged cumulative exposure on outcomes per 1,000 residents of 1987. Columns (1) and (2) use 1992–2007. Column (3) is the preferred specification; its divorce and marriage estimates are those of Table 6. Within a column, all outcomes share the same first stage. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.8: Births by the parents' marital status

	All births (1)	Marital births (2)	Nonmarital births (3)	Nonmarital share (%) (4)
Cumulative exposure $_{t-1}$	0.0269	0.0353	−0.0084	0.0134
	(0.0308)	(0.0286)	(0.0060)	(0.0609)
Kleibergen–Paap F	38.29	38.29	38.29	38.65
Observations	5,120	5,120	5,120	5,120

Notes: 2SLS estimates in the lagged-outcome specification, 1992–2007: state-by-year fixed effects, baseline controls, and the 1987 levels of the birth rate and the nonmarital birth rate (columns 1–3) or of the nonmarital share (column 4). Columns (1)–(3) count live births per 1,000 residents of 1987; columns (2) and (3) sum to column (1). Column (4) is the percentage of live births to unmarried parents. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.9: Placebo: expellees who came to the West directly

	First stage Exposure $_{t-1}$ (1)	Reduced form Divorces (2)	2SLS Divorces (3)
Instrument	0.5656***	−0.0141***	
	(0.1069)	(0.0052)	
Placebo instrument	−0.0243	−0.0012	
	(0.0326)	(0.0017)	
Cumulative exposure $_{t-1}$			−0.0250***
			(0.0094)
Placebo share × year			✓
Kleibergen–Paap F			27.77
Observations	6,400	6,400	6,400

Notes: All columns use the preferred specification: district fixed effects, state-by-year fixed effects and baseline controls interacted with year indicators, on 1985–1988 and 1992–2007. Columns (1) and (2) regress lagged cumulative exposure and annual divorces per 1,000 residents of 1987 on the instrument Z_{it} of equation (10) and the placebo instrument Z^P_{it} of equation (15); without the placebo instrument, the coefficients on Z_{it} are 0.548 and −0.0150 (Tables 3 and 4). Column (3) adds the placebo share s^P_i interacted with year indicators to the preferred 2SLS specification, whose estimate is −0.0273 (Table 7, column 1). s^P_i: expellees from the former eastern territories who came to the Federal Republic without first living in the Soviet zone or Berlin, as a fraction of the 1961 population (Appendix A.3). Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.6 Income growth and the divorce estimate

Table B.10 reports equation (11) with log income per inhabitant as the outcome, estimated on the years in which district income is observed, with 1989 as the reference year; Figure B.3 plots the year-by-year coefficients of the second row of Panel A. Within each vintage, where no adjustment is needed, income in districts with stronger ties grew at essentially the same rate as elsewhere: by 0.22 percent more per percentage point of the share between 1989 and 1992 (s.e. 0.20; column 1, first row), and by 0.21 percent less on average after 1995 (s.e. 0.15; column 4). GDP at the place of work, available from 1992, did not grow faster either: relative to 1992, the mean coefficient over 1995–2007 is −0.0036 (s.e. 0.0040) for GDP per inhabitant and −0.0017 (s.e. 0.0029) for GDP per person employed.

Table B.11 shows that the divorce estimates do not depend on income. Adding year-specific effects of log 1989 income to the preferred specification raises the 2SLS estimate slightly, to −0.0298, leaves the event-study coefficients essentially unchanged, and keeps the pre-reunification coefficients jointly insignificant. Controlling for current log income per inhabitant, which may itself respond to the arrivals, leaves the 2SLS estimate at −0.0271.

The bound in Section 5.3 follows Conley et al. (2012). If the share also affected divorce through income, with elasticity ε, the reduced form would contain the direct effect $\varepsilon\,\bar{y}\,\phi_Z$, and the 2SLS estimate net of it is

$$\beta(\varepsilon) = \frac{\rho - \varepsilon\,\bar{y}\,\phi_Z}{\pi}, \tag{16}$$

where $\rho = -0.0150$ and $\pi = 0.548$ are the reduced-form and first-stage coefficients of the preferred specification (Tables 4 and 3), $\bar{y} = 2.52$ is the mean divorce rate over 1992–2007 (Table 2), and $\phi_Z = -0.0009$ (s.e. 0.0007) is the coefficient on Z_{it} in a regression of log disposable income per inhabitant on Z_{it}, on the years in which income is observed, with the fixed effects and controls of the preferred specification and year-specific effects of log 1989 income. The income channel would account for the entire estimate only at $\varepsilon = \rho/(\bar{y}\,\phi_Z) = 6.7$. The event-study analogue, the mean divorce coefficient over 1995–2007 in Table B.5 divided by $\bar{y}$ times the corresponding income coefficient (column 3 of Panel A, second row, of Table B.10), is 5.3.

B.7 Single states and migration in both directions

Table B.12 re-estimates the preferred specification without one state at a time and without the Bavarian regions on the Czech border (Section 5.2). Table B.13 reports how the instrument relates to arrivals from and departures to the East, the 2SLS estimates for alternative measures of exposure, and net migration with other West German districts

Table B.10: Historical East ties and district income, 1989–2007

	Relative to 1989			Relative to 1995
	1992	1995	Mean 1995–2007	Mean 1996–2007
	(1)	(2)	(3)	(4)
Panel A. Log disposable income per inhabitant				
Preferred specification	0.0022	−0.0058**	−0.0077**	−0.0021
	(0.0020)	(0.0029)	(0.0031)	(0.0015)
+ 1989 income × year	0.0039**	−0.0011	−0.0031	
	(0.0017)	(0.0020)	(0.0021)	
+ border-distance bins × year	0.0027	−0.0024	−0.0037	
	(0.0019)	(0.0021)	(0.0023)	
Panel B. Log primary income per inhabitant				
Preferred specification	0.0002	−0.0071***	−0.0062***	0.0010
	(0.0016)	(0.0022)	(0.0023)	(0.0020)
+ 1989 income × year	0.0020	−0.0035	−0.0030	
	(0.0016)	(0.0027)	(0.0022)	

Notes: Coefficients of the 1961 East-linked share (in percentage points) interacted with year indicators, with district fixed effects, state-by-year fixed effects and baseline controls interacted with year indicators, as in the preferred specification; 320 districts in 1989, 1992 and 1995–2007 (4,800 observations). Columns (1) and (2) report the coefficients for 1992 and 1995 and column (3) the mean of the 1995–2007 coefficients. Column (4) reports the mean of the 1996–2007 coefficients from the same specification estimated on 1995–2007 alone (4,160 observations); within one vintage, no revision needs to be absorbed. Rows marked + add log disposable income per inhabitant in 1989 interacted with year indicators; the third row of Panel A also adds 50-kilometer bins of distance to the former inner-German border interacted with year indicators. A coefficient of 0.01 means 1 percent faster growth per percentage point of the share; the estimates of Burchardi & Hassan (2013) imply about 0.024 for 1989–1995. Data: Appendix A.4. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(Sections 5.1 and 5.4).

Table B.11: Divorce estimates conditional on district income

	Preferred specification (1)	+ 1989 income × year (2)	+ Log income per inhabitant (3)
Panel A. 2SLS			
Cumulative exposure$_{t-1}$	−0.0273***	−0.0298***	−0.0271***
	(0.0094)	(0.0108)	(0.0097)
Kleibergen–Paap F	27.40	27.90	26.74
Observations	6,400	6,400	6,400
Panel B. Event study			
Mean 1995–2007 coefficient	−0.0418***	−0.0390***	
	(0.0136)	(0.0135)	
Joint pre-trend test, 1985–1987 (p)	0.261	0.258	
Observations	7,360	7,360	

Notes: The dependent variable is annual divorces per 1,000 residents of 1987. Column (1) repeats the preferred specification (Table 7, column 1, and Table B.5). Column (2) adds log disposable income per inhabitant in 1989 interacted with year indicators. Column (3) adds current log disposable income per inhabitant as a control, interpolated for 1993 and 1994 and set to its 1989 value for 1985–1988. Panel B estimates equation (11) and reports the mean of the 1995–2007 coefficients and the p-value of the joint test that the 1985–1987 coefficients are zero, estimated on 1985–1988 as in Table B.5. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.12: Excluding one state at a time

Excluded	Districts	2SLS		Kleibergen–Paap F	1961 share × post
		Coefficient	s.e.		
Schleswig-Holstein	305	−0.0299	(0.0105)	22.90	−0.0540 (0.0106)
Lower Saxony	277	−0.0365	(0.0202)	8.07	−0.0598 (0.0133)
North Rhine-Westphalia	266	−0.0268	(0.0081)	30.71	−0.0555 (0.0114)
Hesse	294	−0.0237	(0.0094)	23.03	−0.0523 (0.0110)
Rhineland-Palatinate	284	−0.0228	(0.0087)	27.87	−0.0468 (0.0098)
Baden-Württemberg	276	−0.0279	(0.0098)	26.67	−0.0585 (0.0116)
Bavaria	224	−0.0239	(0.0108)	28.36	−0.0450 (0.0123)
Saarland	314	−0.0279	(0.0095)	27.16	−0.0542 (0.0104)
Czech-border regions of Bavaria	285	−0.0301	(0.0098)	31.44	−0.0539 (0.0108)

Notes: Each row re-estimates the preferred specification for divorces without the districts named in the first column. *2SLS*: coefficient of cumulative exposure in equation (7). *1961 share* × *post*: θ in equation (12). The last row excludes the 35 districts of the administrative regions of Lower Bavaria, Upper Palatinate and Upper Franconia. Standard errors clustered by district in parentheses.

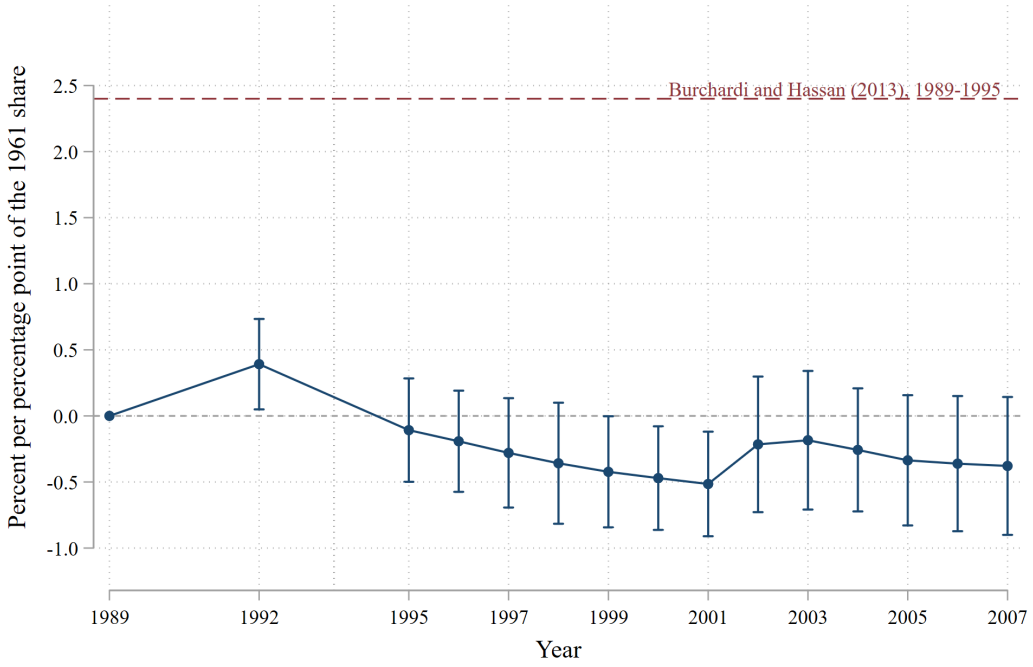


Figure B.3: Disposable income per inhabitant by historical East-linked share, 1989–2007

Notes: Coefficients of the 1961 share (in percentage points) interacted with year indicators, with log disposable income per inhabitant as the outcome and 1989 as the reference year, from the specification in the second row of Table B.10, Panel A, multiplied by 100. Bars show 95% confidence intervals based on standard errors clustered by district. The dotted line marks the change of vintage between 1992 and 1995; the dashed line marks the difference that the estimates of Burchardi & Hassan (2013) imply for 1989–1995, 2.4 percent per percentage point of their tie measure.

Table B.13: Historical ties and migration in both directions

	Coefficient	s.e.	Kleibergen–Paap F
Panel A. Cumulative flows per 1,000 residents on the instrument Z_{it}			
Arrivals from the East, ages 18–29	0.548***	(0.105)	
Arrivals minus departures, ages 18–29	0.205***	(0.057)	
Arrivals from the East, all ages	1.128***	(0.248)	
Departures to the East, all ages	0.833***	(0.261)	
Arrivals minus departures, all ages	0.295	(0.188)	
Departures to the East, ages 30–49	0.226***	(0.082)	
Panel B. 2SLS estimates for divorces by measure of exposure			
Arrivals, ages 18–29 (Table 6)	−0.0273***	(0.0094)	27.40
Arrivals, all ages	−0.0133***	(0.0049)	20.69
Arrivals minus departures, ages 18–29	−0.0730**	(0.0308)	13.03
Arrivals minus departures, all ages	−0.0508	(0.0379)	2.46
Panel C. Annual net migration with other West German districts on exposure, 2SLS			
All ages	0.261	(0.303)	29.01
Ages 30–49	0.064	(0.095)	29.01

Notes: Flows are from the district-to-district migration matrices (Appendix A.2), per 1,000 residents of 1988. Panel A: regressions of the cumulative flow through year $t-1$ on the instrument of equation (10) with the fixed effects and controls of the preferred specification, on 1985–1988 and 1992–2007 (6,400 observations); as for exposure, cumulative flows are zero through 1991. The first row is the first stage of Table 3, column (6). Panel B: equation (7) for divorces with the measure of exposure named in the row. Panel C: annual arrivals from minus departures to other West German districts as the outcome in equation (7), 1992–2007 (5,120 observations); a negative coefficient would mean that districts lose residents to the rest of West Germany as exposure grows. Standard errors clustered by district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.